

Trajectory Uncertainty Calibration for Multimodal VLMs

Complete Benchmark & Transfer LaTeX Table Compendium

M3-LLaVA (7B) & MQT-LLaVA (7B) Evaluations

September 28, 2026

Abstract

This document compiles all 37 publication-ready LaTeX tables across Groups 1–8 and RQ1 Master Tables for local inspection before Overleaf integration. All tables follow the standardized 4-metric schema: ECE (%) ↓, Ada-ECE (%) ↓, Brier score ↓, and AUROC ↑, featuring Option A ranking (**bold** Rank 1, *italic* Rank 2), Core 5 macro aggregation, Trajectory Platt (5D) evaluation, dedicated VQAv2 single-pass vs. multi-rollout comparison, and isolated RQ1 Core 5 and Adversarial master benchmarks.

Contents

1	Main Manuscript Tables (Groups 5, 2, 6, 7, 4, 9)	2
1.1	Table 2: Zero-Shot Cross-Domain Calibration Transfer (M3-LLaVA 7B)	2
1.2	Table 3: Zero-Shot Cross-Domain Calibration Transfer (MQT-LLaVA 7B)	3
1.3	Group 2: Macro-Averaged Calibration Benchmark Across Core 5 Datasets	4
1.4	Comprehensive Core 5 Dataset Benchmark Breakdown (M3 & MQT 7B)	5
1.5	RQ1 Master Table: Core 5 Vision-Language Calibration Benchmark (M3 & MQT 7B)	6
1.6	RQ1 Master Table: Adversarial & Safety Benchmark (M3 & MQT 7B)	7
1.7	Group 9: VQAv2 Single-Pass (1×) vs. Multi-Rollout UMPIRE Suite (10×)	8
1.8	Group 6: Zero-Shot Source Anchor Leaderboard	9
1.9	Group 7: Sample Efficiency & Low-Data Calibration Scaling	10
1.10	Group 4: Leave-One-Dataset-Out (LODO) Cross-Domain Transfer	13
2	Group 3: Temperature Transfer Robustness Ablation	14
2.1	Temperature Transfer: Macro-Averaged Across 4 Benchmarks	14
2.2	Temperature Transfer: POPE	17
2.3	Temperature Transfer: ScienceQA	20
2.4	Temperature Transfer: TextVQA	23
2.5	Temperature Transfer: VizWiz-VQA	26
3	Group 1: Per-Dataset In-Domain Calibration Benchmarks	29
3.1	Benchmark: AI2D	29
3.2	Benchmark: AVQA	30
3.3	Benchmark: CHARTQA	31
3.4	Benchmark: DOCVQA	32
3.5	Benchmark: GQA	33
3.6	Benchmark: INFOGRAPHICVQA	34
3.7	Benchmark: LEGOPUZZLES	35
3.8	Benchmark: MMBENCH	36
3.9	Benchmark: MMMU	37
3.10	Benchmark: POPE	38
3.11	Benchmark: SCIENCEQA	39
3.12	Benchmark: SEEDBENCH	40
3.13	Benchmark: TEXTVQA	41

3.14	Benchmark: VIZWIZVQA	42
3.15	Benchmark: VLLM-SAFETY	43
3.16	Benchmark: VQAV2	44
4	Group 8: Combinatorial 5D Trajectory Feature Ablation Study	45
4.1	Table 4: Macro Cardinality Progression ($k \in \{1 \dots 5\}$)	45
4.2	Table 5: Leave-One-Out (LOO) Feature Degradation Sensitivity	46
4.3	Table 6: Exhaustive 14-Benchmark Adaptive ECE Progression Grid	47
4.4	Table 7: Cardinality Progression on TextVQA	48
4.5	Table 8: Cardinality Progression on VQAv2	49
4.6	Table 9: Cardinality Progression on LegoPuzzles	50

Table 1: **Zero-Shot Cross-Domain Calibration Transfer (M3-LLaVA 7B)**. Pairwise zero-shot calibration transfer from source datasets across held-out target vision-language benchmarks without target domain supervision. Evaluated across Expected Calibration Error (**ECE** ↓), Adaptive ECE (**Ada-ECE** ↓), AUROC (↑), and Brier score (↓). **Bold**: Rank 1; *italic*: Rank 2 among transfer methods.

Source: VQAv2 (General Scene VQA)												
Method	TextVQA (Scene Text)				AI2D (Diagrams)				VizWiz-VQA (Assistive)			
	ECE ↓	Ada-ECE ↓	AUROC ↑	Brier ↓	ECE ↓	Ada-ECE ↓	AUROC ↑	Brier ↓	ECE ↓	Ada-ECE ↓	AUROC ↑	Brier ↓
IN-DOMAIN CALIBRATION (PLATT 5D)	0.0578	0.0535	0.835	0.1628	0.0861	0.1115	0.661	0.2289	0.0638	0.0657	0.794	0.1828
Uncalibrated (NC)	<i>0.0851</i>	<i>0.0867</i>	<i>0.837</i>	0.1717	0.4210	0.4210	0.638	0.4209	0.1703	0.1703	0.789	<i>0.2118</i>
Temperature Scaling (TS)	0.1030	0.1012	<i>0.837</i>	0.1739	0.4201	0.4201	0.638	0.4200	0.1544	0.1554	0.789	0.2099
Adaptive TS (ATS)	0.1154	0.1126	0.840	<i>0.1737</i>	0.4196	0.4196	0.667	0.4194	<i>0.1598</i>	<i>0.1598</i>	<i>0.787</i>	0.2125
Platt Scaling (1D)	0.1788	0.1785	<i>0.837</i>	0.2066	<i>0.4184</i>	<i>0.4184</i>	0.638	<i>0.4183</i>	0.2765	0.2765	0.789	0.2741
Platt 5D (Ours)	0.0736	0.0754	0.811	0.1793	0.4026	0.4026	<i>0.661</i>	0.4027	0.2400	0.2400	0.716	0.2677
Source: VizWiz-VQA (Assistive / Blurry)												
Method	TextVQA (Scene Text)				AI2D (Diagrams)				VQAv2 (General Scene VQA)			
	ECE ↓	Ada-ECE ↓	AUROC ↑	Brier ↓	ECE ↓	Ada-ECE ↓	AUROC ↑	Brier ↓	ECE ↓	Ada-ECE ↓	AUROC ↑	Brier ↓
IN-DOMAIN CALIBRATION (PLATT 5D)	0.0578	0.0535	0.835	0.1628	0.0861	0.1115	0.661	0.2289	0.0686	0.0650	0.788	0.1380
Uncalibrated (NC)	0.0851	0.0867	0.837	0.1717	0.4210	0.4210	0.638	0.4209	0.0791	0.0769	<i>0.764</i>	0.1370
Temperature Scaling (TS)	0.0886	<i>0.0851</i>	0.837	0.1717	0.4207	0.4207	0.638	0.4206	<i>0.0892</i>	<i>0.0915</i>	<i>0.764</i>	<i>0.1407</i>
Adaptive TS (ATS)	0.0853	0.0854	<i>0.835</i>	0.1720	0.4207	0.4207	0.605	<i>0.4207</i>	0.0915	0.0916	0.760	0.1414
Platt Scaling (1D)	<i>0.0842</i>	0.0896	0.837	<i>0.1689</i>	<i>0.4209</i>	<i>0.4209</i>	0.638	0.4208	0.1779	0.1782	<i>0.764</i>	0.1767
Platt 5D (Ours)	0.0776	0.0779	<i>0.835</i>	0.1683	<i>0.4209</i>	<i>0.4209</i>	<i>0.637</i>	0.4208	0.1788	0.1776	0.765	0.1772
Source: TextVQA (Scene Text)												
Method	VQAv2 (General Scene VQA)				VizWiz-VQA (Assistive)				AI2D (Diagrams)			
	ECE ↓	Ada-ECE ↓	AUROC ↑	Brier ↓	ECE ↓	Ada-ECE ↓	AUROC ↑	Brier ↓	ECE ↓	Ada-ECE ↓	AUROC ↑	Brier ↓
IN-DOMAIN CALIBRATION (PLATT 5D)	0.0686	0.0650	0.788	0.1380	0.0638	0.0657	0.794	0.1828	0.0861	0.1115	0.661	0.2289
Uncalibrated (NC)	0.0791	<i>0.0769</i>	0.764	0.1370	0.1703	0.1703	0.789	0.2118	<i>0.4210</i>	<i>0.4210</i>	0.638	<i>0.4209</i>
Temperature Scaling (TS)	0.0811	0.0799	0.764	0.1380	0.1660	0.1660	0.789	0.2102	0.4209	0.4209	0.638	<i>0.4209</i>
Adaptive TS (ATS)	<i>0.0799</i>	0.0757	<i>0.768</i>	<i>0.1372</i>	0.1689	0.1689	<i>0.788</i>	0.2112	0.4209	0.4209	0.658	<i>0.4209</i>
Platt Scaling (1D)	0.1235	0.1241	0.764	0.1507	0.0885	0.0873	0.789	0.1881	0.4209	0.4209	0.638	<i>0.4209</i>
Platt 5D (Ours)	0.0903	0.0928	0.770	0.1417	<i>0.1192</i>	<i>0.1192</i>	0.780	<i>0.1988</i>	0.4209	0.4209	<i>0.648</i>	0.4208

1 Main Manuscript Tables (Groups 5, 2, 6, 7, 4, 9)

1.1 Table 2: Zero-Shot Cross-Domain Calibration Transfer (M3-LLaVA 7B)

Table 2: **Zero-Shot Cross-Domain Calibration Transfer (MQT-LLaVA 7B)**. Pairwise zero-shot calibration transfer from source datasets across held-out target vision-language benchmarks without target domain supervision. Evaluated across Expected Calibration Error (**ECE** ↓), Adaptive ECE (**Ada-ECE** ↓), AUROC (↑), and Brier score (↓). **Bold**: Rank 1; *italic*: Rank 2 among transfer methods.

Source: VQAv2 (General Scene VQA)												
Method	TextVQA (Scene Text)				AI2D (Diagrams)				VizWiz-VQA (Assistive)			
	ECE ↓	Ada-ECE ↓	AUROC ↑	Brier ↓	ECE ↓	Ada-ECE ↓	AUROC ↑	Brier ↓	ECE ↓	Ada-ECE ↓	AUROC ↑	Brier ↓
IN-DOMAIN CALIBRATION (PLATT 5D)	0.0525	0.0776	0.719	0.1948	0.0627	0.0653	0.739	0.2035	0.0603	0.0817	0.795	0.1418
Uncalibrated (NC)	0.3168	0.3168	0.722	0.3096	0.2631	0.2625	0.733	0.2909	0.3203	0.3203	<i>0.809</i>	0.2488
Temperature Scaling (TS)	<i>0.2757</i>	<i>0.2757</i>	0.722	0.2753	0.1745	0.1743	0.733	0.2387	0.3053	0.3053	<i>0.809</i>	0.2343
Adaptive TS (ATS)	0.2758	0.2758	0.740	<i>0.2705</i>	<i>0.1818</i>	<i>0.1818</i>	<i>0.729</i>	<i>0.2423</i>	<i>0.3148</i>	<i>0.3148</i>	0.811	<i>0.2386</i>
Platt Scaling (1D)	0.3714	0.3714	0.722	0.3388	0.2362	0.2362	0.733	0.2718	0.4387	0.4385	<i>0.809</i>	0.3430
Platt 5D (Ours)	0.2356	0.2357	<i>0.730</i>	0.2569	0.2130	0.2112	0.648	0.2761	0.3475	0.3470	0.745	0.2798
Source: VizWiz-VQA (Assistive / Blurry)												
Method	TextVQA (Scene Text)				AI2D (Diagrams)				VQAv2 (General Scene VQA)			
	ECE ↓	Ada-ECE ↓	AUROC ↑	Brier ↓	ECE ↓	Ada-ECE ↓	AUROC ↑	Brier ↓	ECE ↓	Ada-ECE ↓	AUROC ↑	Brier ↓
IN-DOMAIN CALIBRATION (PLATT 5D)	0.0525	0.0776	0.719	0.1948	0.0627	0.0653	0.739	0.2035	0.0716	0.0764	0.739	0.1629
Uncalibrated (NC)	0.3168	0.3168	<i>0.722</i>	0.3096	0.2631	0.2625	<i>0.733</i>	0.2909	0.0745	0.0748	<i>0.720</i>	0.1568
Temperature Scaling (TS)	0.2565	0.2565	<i>0.722</i>	0.2654	0.1424	0.1415	<i>0.733</i>	0.2283	<i>0.1371</i>	<i>0.1381</i>	<i>0.720</i>	<i>0.1727</i>
Adaptive TS (ATS)	0.2588	0.2588	0.714	0.2686	0.1448	0.1438	0.735	0.2287	0.1486	0.1444	0.715	0.1739
Platt Scaling (1D)	<i>0.1475</i>	<i>0.1447</i>	<i>0.722</i>	<i>0.2331</i>	<i>0.1121</i>	<i>0.1131</i>	<i>0.733</i>	0.2190	0.3261	0.3266	<i>0.720</i>	0.2837
Platt 5D (Ours)	0.1340	0.1316	0.736	0.2185	0.1026	0.1058	0.727	<i>0.2219</i>	0.3148	0.3144	0.730	0.2724
Source: TextVQA (Scene Text)												
Method	VQAv2 (General Scene VQA)				VizWiz-VQA (Assistive)				AI2D (Diagrams)			
	ECE ↓	Ada-ECE ↓	AUROC ↑	Brier ↓	ECE ↓	Ada-ECE ↓	AUROC ↑	Brier ↓	ECE ↓	Ada-ECE ↓	AUROC ↑	Brier ↓
IN-DOMAIN CALIBRATION (PLATT 5D)	0.0716	0.0764	0.739	0.1629	0.0603	0.0817	0.795	0.1418	0.0627	0.0653	0.739	0.2035
Uncalibrated (NC)	0.0745	0.0748	0.720	0.1568	0.3203	0.3203	0.809	0.2488	0.2631	0.2625	0.733	0.2909
Temperature Scaling (TS)	0.2230	0.2409	0.720	0.2190	0.2855	0.2926	0.809	0.2435	0.1269	0.1543	0.733	0.2299
Adaptive TS (ATS)	<i>0.2200</i>	<i>0.2374</i>	<i>0.723</i>	<i>0.2167</i>	0.2830	0.2936	0.809	0.2433	0.1213	0.1499	0.733	<i>0.2289</i>
Platt Scaling (1D)	0.3649	0.3632	0.720	0.3048	<i>0.1028</i>	<i>0.1179</i>	0.809	<i>0.1521</i>	<i>0.1219</i>	0.1219	0.733	0.2203
Platt 5D (Ours)	0.2881	0.2881	0.755	0.2402	0.0868	0.0842	<i>0.765</i>	0.1501	0.1349	<i>0.1364</i>	<i>0.648</i>	0.2535

1.2 Table 3: Zero-Shot Cross-Domain Calibration Transfer (MQT-LLaVA 7B)

Table 3: **Macro-Averaged Calibration Benchmark Across Core 5 Datasets (M3-LLaVA 7B).** **Bold:** best; *italic:* second best. ↓/↑: lower/higher is better.

Calibration Method	Regime / Sampling	Macro ECE (%) ↓	Macro Ada-ECE (%) ↓	Macro Brier ↓	Macro AUROC ↑
Naive Confidence (NC)	Single-Pass ($T = 0.0$)	19.85	19.64	0.2348	0.765
Temperature Scaling (TS)	Single-Pass ($T = 0.0$)	11.32	11.94	0.1919	0.765
Adaptive TS (ATS)	Single-Pass ($T = 0.0$)	11.01	11.97	0.1909	<i>0.771</i>
Platt Scaling (1D)	Single-Pass ($T = 0.0$)	6.21	6.73	<i>0.1788</i>	0.765
Trajectory Platt (5D)	Single-Pass ($T = 0.0$)	<i>6.56</i>	<i>7.15</i>	0.1769	0.778

Table 4: **Macro-Averaged Calibration Benchmark Across Core 5 Datasets (MQT-LLaVA 7B).** **Bold:** best; *italic:* second best. ↓/↑: lower/higher is better.

Calibration Method	Regime / Sampling	Macro ECE (%) ↓	Macro Ada-ECE (%) ↓	Macro Brier ↓	Macro AUROC ↑
Naive Confidence (NC)	Single-Pass ($T = 0.0$)	24.84	24.60	0.2580	0.726
Temperature Scaling (TS)	Single-Pass ($T = 0.0$)	16.82	16.94	0.2158	0.726
Adaptive TS (ATS)	Single-Pass ($T = 0.0$)	16.39	16.59	0.2145	<i>0.730</i>
Platt Scaling (1D)	Single-Pass ($T = 0.0$)	<i>7.93</i>	<i>8.84</i>	<i>0.1845</i>	0.726
Trajectory Platt (5D)	Single-Pass ($T = 0.0$)	6.24	7.16	0.1778	0.752

1.3 Group 2: Macro-Averaged Calibration Benchmark Across Core 5 Datasets

Table 5: **Comprehensive Calibration Benchmark Breakdown Across Core 5 Datasets.** Evaluated across Adaptive ECE (Ada-ECE (%) ↓) and AUROC (↑) on M3-LLaVA (7B) (Panel A) and MQT-LLaVA (7B) (Panel B). **Bold:** best; *italic:* second best within each metric column. Clean unshaded presentation.

Calibration Method	AI2D		ScienceQA		TextVQA		VizWiz-VQA		VQAv2		Average	
	Ada ↓	AUC ↑	Ada ↓	AUC ↑	Ada ↓	AUC ↑	Ada ↓	AUC ↑	Ada ↓	AUC ↑	Ada ↓	AUC ↑
Panel A: M3-LLaVA (7B)												
Naive Confidence (NC)	42.00	0.614	21.93	0.807	11.07	0.839	15.97	<i>0.793</i>	7.23	<i>0.772</i>	19.64	0.765
Temperature Scaling (TS)	12.18	0.614	12.68	0.807	11.14	0.839	15.09	<i>0.793</i>	8.58	<i>0.772</i>	11.94	0.765
Adaptive TS (ATS)	13.15	<i>0.625</i>	13.00	<i>0.808</i>	11.31	0.839	14.87	<i>0.793</i>	7.52	0.788	11.97	<i>0.771</i>
Platt Scaling (1D)	7.70	0.614	4.99	0.807	<i>8.22</i>	0.839	5.80	<i>0.793</i>	<i>6.97</i>	<i>0.772</i>	6.73	0.765
Trajectory Platt (5D)	<i>11.15</i>	0.661	<i>6.15</i>	0.813	5.35	<i>0.835</i>	<i>6.57</i>	0.794	6.50	0.788	<i>7.15</i>	0.778
Panel B: MQT-LLaVA (7B)												
Naive Confidence (NC)	26.92	<i>0.746</i>	23.17	0.760	31.91	0.684	30.66	0.797	<i>10.31</i>	0.641	24.60	0.726
Temperature Scaling (TS)	13.10	<i>0.746</i>	10.23	0.760	18.32	0.684	29.41	0.797	13.65	0.641	16.94	0.726
Adaptive TS (ATS)	14.08	0.747	9.38	<i>0.761</i>	18.36	<i>0.689</i>	29.11	<i>0.796</i>	12.03	<i>0.658</i>	16.59	<i>0.730</i>
Platt Scaling (1D)	<i>8.25</i>	<i>0.746</i>	<i>6.91</i>	0.760	<i>10.46</i>	0.684	7.62	0.797	10.94	0.641	<i>8.84</i>	0.726
Trajectory Platt (5D)	6.53	0.739	5.72	0.769	7.76	0.719	<i>8.17</i>	0.795	7.64	0.739	7.16	0.752

1.4 Comprehensive Core 5 Dataset Benchmark Breakdown (M3 & MQT 7B)

Table 6: **RQ1 Evaluation on M3-LLaVA (7B)**. Comparison of calibration methods across 5 vision-language benchmarks. Metrics: Expected Calibration Error (ECE (%) ↓), Adaptive ECE (Ada-ECE (%) ↓), Brier Score (↓), and AUROC (↑). **Bold**: Rank 1; *italic*: Rank 2 among single-pass methods.

Dataset	Naive Confidence				Temperature Scaling				Adaptive TS (ATS)				Platt Scaling (1D)				Trajectory Platt (5D)			
	ECE	Ada.	Brier	AUROC	ECE	Ada.	Brier	AUROC	ECE	Ada.	Brier	AUROC	ECE	Ada.	Brier	AUROC	ECE	Ada.	Brier	AUROC
AI2D	42.00	42.00	0.4200	0.614	8.62	12.18	0.247	0.614	8.64	13.15	0.247	0.625	5.63	7.70	0.237	0.614	8.61	11.15	0.228	0.661
ScienceQA	22.16	21.93	0.236	0.807	13.10	12.68	0.192	0.807	12.98	13.00	0.191	0.808	4.80	4.99	0.173	0.807	5.17	6.15	0.171	0.813
TextVQA	10.50	11.07	0.171	0.839	0.38	11.14	0.171	0.839	10.15	11.31	0.171	0.839	8.47	8.22	0.161	0.839	5.78	5.35	0.162	0.835
VizWiz-VQA	15.97	15.97	0.207	0.793	15.09	15.09	0.206	0.793	14.87	14.87	0.205	0.793	6.64	5.80	0.182	0.793	6.38	6.57	0.182	0.794
VQAv2	8.64	7.23	0.138	0.772	9.40	8.58	0.141	0.772	8.39	7.52	0.138	0.788	5.52	6.97	0.139	0.772	6.86	6.50	0.138	0.788
Macro Avg.	19.85	19.64	0.234	0.765	11.32	11.94	0.191	0.765	11.01	11.97	0.190	0.771	6.21	6.73	0.178	0.765	6.56	7.15	0.176	0.778

Table 7: **RQ1 Evaluation on MQT-LLaVA (7B)**. Comparison of calibration methods across 5 vision-language benchmarks. Metrics: Expected Calibration Error (ECE (%) ↓), Adaptive ECE (Ada-ECE (%) ↓), Brier Score (↓), and AUROC (↑). **Bold**: Rank 1; *italic*: Rank 2 among single-pass methods.

Dataset	Naive Confidence				Temperature Scaling				Adaptive TS (ATS)				Platt Scaling (1D)				Trajectory Platt (5D)			
	ECE	Ada.	Brier	AUROC	ECE	Ada.	Brier	AUROC	ECE	Ada.	Brier	AUROC	ECE	Ada.	Brier	AUROC	ECE	Ada.	Brier	AUROC
AI2D	27.15	26.92	0.288	0.746	13.31	13.10	0.219	0.746	13.38	14.08	0.219	0.747	5.92	8.25	0.201	0.746	6.27	6.53	0.203	0.739
ScienceQA	23.65	23.17	0.257	0.760	9.32	10.23	0.197	0.760	8.73	9.38	0.196	0.761	7.99	6.91	0.191	0.760	6.46	5.72	0.186	0.769
TextVQA	32.27	31.91	0.328	0.684	18.21	18.32	0.250	0.684	18.36	18.36	0.250	0.689	8.95	10.46	0.214	0.684	5.25	7.76	0.194	0.719
VizWiz-VQA	30.66	30.66	0.243	0.797	29.41	29.41	0.232	0.797	29.11	29.11	0.231	0.796	6.27	7.62	0.143	0.797	6.03	8.17	0.141	0.795
VQAv2	10.49	10.31	0.171	0.641	13.87	13.65	0.178	0.641	12.36	12.03	0.175	0.658	10.54	10.94	0.171	0.641	7.16	7.64	0.162	0.739
Macro Avg.	24.84	24.60	0.258	0.726	16.82	16.94	0.215	0.726	16.39	16.59	0.214	0.730	7.93	8.84	0.184	0.726	6.24	7.16	0.177	0.752

1.5 RQ1 Master Table: Core 5 Vision-Language Calibration Benchmark (M3 & MQT 7B)

tabularx array

Table 8: **RQ1 Evaluation: Calibration Robustness Under Adversarial and Safety Stress Benchmarks.** Evaluated across Expected Calibration Error (ECE (%) ↓), Adaptive ECE (Ada-ECE (%) ↓), Brier Score (↓), and AUROC (↑) on M3-LLaVA (7B) (Panel A) and MQT-LLaVA (7B) (Panel B) across AVQA, VLLM-Safety, and Adversarial Macro Average. **Bold:** Rank 1; *italic:* Rank 2 within each metric column among single-pass methods. Multi-pass baselines report placeholders (–) pending rerun completion. Clean unshaded presentation.

Calibration Method	AVQA				VLLM-Safety				Macro Average			
	ECE (%) ↓	Ada-ECE (%) ↓	Brier ↓	AUROC ↑	ECE (%) ↓	Ada-ECE (%) ↓	Brier ↓	AUROC ↑	ECE (%) ↓	Ada-ECE (%) ↓	Brier ↓	AUROC ↑
Panel A: M3-LLaVA (7B)												
Naive Confidence (NC)	25.66	25.40	0.3194	0.599	<i>11.59</i>	<i>10.50</i>	0.1732	0.814	18.63	17.95	0.2463	<i>0.707</i>
Temperature Scaling (TS)	<i>4.45</i>	17.96	0.2480	0.599	16.98	16.80	0.1812	0.814	10.71	17.38	0.2146	<i>0.707</i>
Adaptive TS (ATS)	4.29	<i>16.94</i>	0.2477	<i>0.601</i>	14.08	13.84	<i>0.1730</i>	<i>0.813</i>	<i>9.18</i>	<i>15.39</i>	<i>0.2104</i>	<i>0.707</i>
Platt Scaling (1D)	6.84	17.67	<i>0.2463</i>	0.599	14.76	14.04	0.1798	0.814	10.80	15.85	0.2131	<i>0.707</i>
LN-Entropy	-	-	-	-	-	-	-	-	-	-	-	-
Semantic Entropy	-	-	-	-	-	-	-	-	-	-	-	-
EigenScore	-	-	-	-	-	-	-	-	-	-	-	-
UMPIRE	-	-	-	-	-	-	-	-	-	-	-	-
Trajectory Platt (5D)	5.26	7.45	0.2240	0.691	5.87	6.88	0.1709	0.773	5.57	7.17	0.1975	0.732
Panel B: MQT-LLaVA (7B)												
Naive Confidence (NC)	27.88	28.00	0.3385	0.602	<i>9.63</i>	<i>9.74</i>	0.1966	<i>0.763</i>	18.76	18.87	0.2675	0.682
Temperature Scaling (TS)	6.56	13.28	0.2491	0.602	14.20	14.42	0.2054	<i>0.763</i>	10.38	13.85	0.2272	0.682
Adaptive TS (ATS)	6.62	<i>12.86</i>	0.2488	<i>0.604</i>	12.13	13.19	0.1966	0.768	9.37	13.03	0.2227	<i>0.686</i>
Platt Scaling (1D)	1.18	13.66	<i>0.2449</i>	0.602	11.34	11.29	<i>0.1937</i>	<i>0.763</i>	<i>6.26</i>	<i>12.48</i>	<i>0.2193</i>	0.682
LN-Entropy	-	-	-	-	-	-	-	-	-	-	-	-
Semantic Entropy	-	-	-	-	-	-	-	-	-	-	-	-
EigenScore	-	-	-	-	-	-	-	-	-	-	-	-
UMPIRE	-	-	-	-	-	-	-	-	-	-	-	-
Trajectory Platt (5D)	<i>5.02</i>	6.54	0.2332	0.654	5.68	6.58	0.1921	0.733	5.35	6.56	0.2127	0.694

1.6 RQ1 Master Table: Adversarial & Safety Benchmark (M3 & MQT 7B)

Table 9: **VQAv2 Calibration: Single-Pass Trajectory Calibration (1×) vs. Multi-Rollout UMPIRE Suite (10×) (M3-LLaVA 7B)**. Bold: best; *italic*: second best. ↓/↑: lower/higher is better.

Calibration Method	Regime / Sampling	ECE (%) ↓	Ada-ECE (%) ↓	Brier ↓	AUROC ↑
Naive Confidence (NC)	Single-Pass ($T = 0.0, K = 1$)	8.64	7.23	0.1387	0.772
Temperature Scaling (TS)	Single-Pass ($T = 0.0, K = 1$)	9.40	8.58	0.1417	0.772
Adaptive TS (ATS)	Single-Pass ($T = 0.0, K = 1$)	8.39	7.52	<i>0.1381</i>	0.788
Platt Scaling (1D)	Single-Pass ($T = 0.0, K = 1$)	5.52	<i>6.97</i>	0.1391	0.772
LN-Entropy	Multi-Pass ($T = 0.5, K = 10$)	12.40	-	-	0.787
Semantic Entropy	Multi-Pass ($T = 0.5, K = 10$)	11.40	-	-	0.690
EigenScore	Multi-Pass ($T = 0.5, K = 10$)	6.20	-	-	<i>0.816</i>
UMPIRE	Multi-Pass ($T = 0.5, K = 10$)	<i>5.80</i>	-	-	0.819
Trajectory Platt (5D)	Single-Pass ($T = 0.0, K = 1$)	6.86	6.50	0.1380	0.788

Table 10: **VQAv2 Calibration: Single-Pass Trajectory Calibration (1×) vs. Multi-Rollout UMPIRE Suite (10×) (MQT-LLaVA 7B)**. Bold: best; *italic*: second best. ↓/↑: lower/higher is better.

Calibration Method	Regime / Sampling	ECE (%) ↓	Ada-ECE (%) ↓	Brier ↓	AUROC ↑
Naive Confidence (NC)	Single-Pass ($T = 0.0, K = 1$)	10.49	<i>10.31</i>	0.1713	0.641
Temperature Scaling (TS)	Single-Pass ($T = 0.0, K = 1$)	13.87	13.65	0.1789	0.641
Adaptive TS (ATS)	Single-Pass ($T = 0.0, K = 1$)	12.36	12.03	0.1753	0.658
Platt Scaling (1D)	Single-Pass ($T = 0.0, K = 1$)	10.54	10.94	<i>0.1712</i>	0.641
LN-Entropy	Multi-Pass ($T = 0.5, K = 10$)	12.30	-	-	0.767
Semantic Entropy	Multi-Pass ($T = 0.5, K = 10$)	13.70	-	-	0.699
EigenScore	Multi-Pass ($T = 0.5, K = 10$)	10.20	-	-	<i>0.773</i>
UMPIRE	Multi-Pass ($T = 0.5, K = 10$)	<i>7.80</i>	-	-	0.790
Trajectory Platt (5D)	Single-Pass ($T = 0.0, K = 1$)	7.16	7.64	0.1629	0.739

1.7 Group 9: VQAv2 Single-Pass (1×) vs. Multi-Rollout UMPIRE Suite (10×)

Table 11: **Zero-Shot Cross-Domain Source Anchor Leaderboard.** Performance of Trajectory Platt (5D) across candidate source anchors vs. standard transfer baselines across 12 unseen target benchmarks. Evaluated zero-shot on M3-LLaVA (7B) and MQT-LLaVA (7B). **Bold:** Rank 1; *Italic:* Rank 2 among transfer configurations. ↓: lower is better; ↑: higher is better.

Configuration / Source Anchor	M3-LLaVA (7B)				MQT-LLaVA (7B)				Dual-Arch
	ECE (%) ↓	Ada-ECE (%) ↓	AUROC ↑	Brier ↓	ECE (%) ↓	Ada-ECE (%) ↓	AUROC ↑	Brier ↓	Total Wins ↑
IN-DOMAIN CALIBRATION (PLATT 5D)	4.77	5.85	0.720	0.1356	5.53	6.42	0.743	0.1312	—
<i>Zero-Shot Transfer Baselines (Macro Mean Across All Pairs)</i>									
Naive Confidence (NC)	41.57	41.55	0.682	0.4076	34.23	34.18	<i>0.688</i>	0.3228	—
Temperature Scaling (TS)	33.27	34.78	0.682	<i>0.3102</i>	30.10	30.46	<i>0.688</i>	0.2581	—
Adaptive TS (ATS)	33.28	34.75	<i>0.686</i>	0.3094	30.11	30.46	0.689	0.2580	—
Platt Scaling (1D)	36.88	37.14	0.649	0.3531	31.42	31.51	0.594	0.2946	—
<i>Trajectory Platt (5D) Candidate Source Anchors</i>									
VQA_{v2}	40.53	40.55	0.687	0.3766	33.81	33.57	0.662	0.3020	23/24
VizWiz-VQA	40.46	40.48	0.671	0.3867	<i>20.96</i>	<i>20.88</i>	0.685	<i>0.2090</i>	<i>20/24</i>
DocVQA	<i>30.03</i>	<i>29.99</i>	0.683	0.3174	37.88	37.89	0.644	0.3686	18/24
TextVQA	42.01	41.99	0.676	0.4052	20.68	20.82	0.668	0.1868	19/24
ChartQA	29.72	29.89	0.672	0.3175	36.09	36.10	0.475	0.3557	16/24
POPE	43.80	43.82	0.677	0.4231	34.29	34.17	0.672	0.3221	9/24

1.8 Group 6: Zero-Shot Source Anchor Leaderboard

1.9 Group 7: Sample Efficiency & Low-Data Calibration Scaling

Table 12: **Calibration Sample Efficiency Scaling Across Training Budgets (M3-LLaVA 7B).** Evaluated across 7 training budgets $N \in \{50, 100, 200, 500, 1000, 1500, \text{Full}\}$ over 5 subsampling seeds. Panel A: TextVQA (Scene Text VQA); Panel B: VQAv2 (General Scene VQA). **Bold**: best; *italic*: second best within each budget tier.

Budget (N)	Calibration Method	ECE (%) ↓	Ada-ECE (%) ↓	AUROC ↑	Brier ↓
Panel A: TextVQA (Scene Text VQA)					
$N = 50$	Naive Confidence (NC)	10.50	11.07	0.839	0.1714
	Temperature Scaling (TS)	11.41	11.43	0.839	0.1726
	Adaptive TS (ATS)	11.89	11.42	0.839	0.1725
	Platt Scaling (1D)	<i>8.73</i>	<i>8.00</i>	0.839	0.1621
	Trajectory Platt (5D)	7.17	7.43	<i>0.817</i>	<i>0.1711</i>
$N = 100$	Naive Confidence (NC)	10.50	11.07	0.839	0.1714
	Temperature Scaling (TS)	10.89	11.30	0.839	0.1719
	Adaptive TS (ATS)	11.14	10.93	0.839	0.1718
	Platt Scaling (1D)	<i>8.75</i>	<i>8.38</i>	0.839	0.1628
	Trajectory Platt (5D)	7.64	7.37	<i>0.829</i>	<i>0.1661</i>
$N = 200$	Naive Confidence (NC)	10.50	11.07	0.839	0.1714
	Temperature Scaling (TS)	10.89	11.37	0.839	0.1722
	Adaptive TS (ATS)	11.28	11.22	0.839	0.1720
	Platt Scaling (1D)	<i>9.46</i>	<i>9.08</i>	0.839	0.1648
	Trajectory Platt (5D)	6.61	7.00	<i>0.824</i>	<i>0.1691</i>
$N = 500$	Naive Confidence (NC)	10.50	11.07	0.839	0.1714
	Temperature Scaling (TS)	10.68	11.18	0.839	0.1714
	Adaptive TS (ATS)	10.73	11.15	0.839	0.1712
	Platt Scaling (1D)	<i>8.35</i>	<i>8.32</i>	0.839	0.1623
	Trajectory Platt (5D)	6.52	5.85	<i>0.832</i>	<i>0.1641</i>
$N = 1000$	Naive Confidence (NC)	10.50	11.07	0.839	0.1714
	Temperature Scaling (TS)	10.12	11.14	0.839	0.1712
	Adaptive TS (ATS)	10.15	11.41	0.839	0.1710
	Platt Scaling (1D)	<i>8.49</i>	<i>8.18</i>	0.839	0.1615
	Trajectory Platt (5D)	6.24	6.56	<i>0.833</i>	<i>0.1636</i>
$N = 1500$	Naive Confidence (NC)	10.50	11.07	0.839	0.1714
	Temperature Scaling (TS)	10.16	11.14	0.839	0.1712
	Adaptive TS (ATS)	9.91	11.31	0.839	0.1710
	Platt Scaling (1D)	<i>8.53</i>	<i>8.24</i>	0.839	0.1616
	Trajectory Platt (5D)	5.63	5.70	<i>0.834</i>	<i>0.1629</i>
Full	Naive Confidence (NC)	10.50	11.07	0.839	0.1714
	Temperature Scaling (TS)	10.38	11.14	0.839	0.1712
	Adaptive TS (ATS)	10.15	11.31	0.839	0.1710
	Platt Scaling (1D)	<i>8.47</i>	<i>8.22</i>	0.839	0.1615
	Trajectory Platt (5D)	5.78	5.35	<i>0.835</i>	<i>0.1628</i>
Panel B: VQAv2 (General Scene VQA)					
$N = 50$	Naive Confidence (NC)	8.64	<i>7.23</i>	<i>0.772</i>	0.1387
	Temperature Scaling (TS)	10.52	9.51	<i>0.772</i>	0.1435
	Adaptive TS (ATS)	8.84	8.29	0.788	<i>0.1394</i>
	Platt Scaling (1D)	6.91	7.90	<i>0.772</i>	0.1411
	Trajectory Platt (5D)	<i>7.14</i>	7.14	0.764	0.1444
$N = 100$	Naive Confidence (NC)	8.64	<i>7.23</i>	0.772	0.1387
	Temperature Scaling (TS)	10.06	9.10	0.772	0.1429

Table 13: **Calibration Sample Efficiency Scaling Across Training Budgets (MQT-LLaVA 7B).** Evaluated across 7 training budgets $N \in \{50, 100, 200, 500, 1000, 1500, \text{Full}\}$ over 5 subsampling seeds. Panel A: TextVQA (Scene Text VQA); Panel B: VQAv2 (General Scene VQA). **Bold**: best; *italic*: second best within each budget tier.

Budget (N)	Calibration Method	ECE (%) ↓	Ada-ECE (%) ↓	AUROC ↑	Brier ↓
Panel A: TextVQA (Scene Text VQA)					
$N = 50$	Naive Confidence (NC)	32.27	31.91	0.684	0.3286
	Temperature Scaling (TS)	19.93	19.96	0.684	0.2539
	Adaptive TS (ATS)	20.59	20.77	<i>0.689</i>	0.2549
	Platt Scaling (1D)	<i>11.84</i>	<i>12.07</i>	0.684	<i>0.2232</i>
	Trajectory Platt (5D)	10.26	10.71	0.694	0.2097
$N = 100$	Naive Confidence (NC)	32.27	31.91	0.684	0.3286
	Temperature Scaling (TS)	19.63	19.68	0.684	0.2534
	Adaptive TS (ATS)	19.85	19.85	<i>0.687</i>	0.2535
	Platt Scaling (1D)	<i>10.34</i>	<i>10.82</i>	0.684	<i>0.2184</i>
	Trajectory Platt (5D)	6.96	7.90	0.705	0.2018
$N = 200$	Naive Confidence (NC)	32.27	31.91	0.684	0.3286
	Temperature Scaling (TS)	18.60	18.67	0.684	0.2509
	Adaptive TS (ATS)	18.73	18.73	<i>0.688</i>	0.2507
	Platt Scaling (1D)	<i>10.01</i>	<i>10.58</i>	0.684	<i>0.2162</i>
	Trajectory Platt (5D)	6.84	7.21	0.711	0.1998
$N = 500$	Naive Confidence (NC)	32.27	31.91	0.684	0.3286
	Temperature Scaling (TS)	18.94	18.95	0.684	0.2514
	Adaptive TS (ATS)	19.14	19.14	<i>0.690</i>	0.2511
	Platt Scaling (1D)	<i>10.58</i>	<i>10.78</i>	0.684	<i>0.2170</i>
	Trajectory Platt (5D)	6.61	8.46	0.717	0.1962
$N = 1000$	Naive Confidence (NC)	32.27	31.91	0.684	0.3286
	Temperature Scaling (TS)	18.59	18.66	0.684	0.2509
	Adaptive TS (ATS)	18.75	18.75	<i>0.689</i>	0.2507
	Platt Scaling (1D)	<i>9.75</i>	<i>10.58</i>	0.684	<i>0.2158</i>
	Trajectory Platt (5D)	5.90	6.97	0.718	0.1956
$N = 1500$	Naive Confidence (NC)	32.27	31.91	0.684	0.3286
	Temperature Scaling (TS)	18.25	18.35	0.684	0.2503
	Adaptive TS (ATS)	18.40	18.40	<i>0.689</i>	0.2501
	Platt Scaling (1D)	<i>8.81</i>	<i>10.45</i>	0.684	<i>0.2148</i>
	Trajectory Platt (5D)	5.51	7.39	0.719	0.1947
Full	Naive Confidence (NC)	32.27	31.91	0.684	0.3286
	Temperature Scaling (TS)	18.21	18.32	0.684	0.2503
	Adaptive TS (ATS)	18.36	18.36	<i>0.689</i>	0.2500
	Platt Scaling (1D)	<i>8.95</i>	<i>10.46</i>	0.684	<i>0.2148</i>
	Trajectory Platt (5D)	5.25	7.76	0.719	0.1948
Panel B: VQAv2 (General Scene VQA)					
$N = 50$	Naive Confidence (NC)	<i>10.49</i>	<i>10.31</i>	0.641	0.1713
	Temperature Scaling (TS)	14.25	13.94	0.641	0.1796
	Adaptive TS (ATS)	12.13	11.33	<i>0.666</i>	0.1735
	Platt Scaling (1D)	10.76	11.61	0.641	0.1723
	Trajectory Platt (5D)	8.40	9.39	0.706	<i>0.1722</i>
$N = 100$	Naive Confidence (NC)	<i>13.49</i>	<i>10.31</i>	0.641	0.1713
	Temperature Scaling (TS)	13.86	13.67	0.641	0.1785

Table 14: **Leave-One-Dataset-Out (LODO) Cross-Domain Transfer (M3-LLaVA 7B)**. Macro-averaged across Core 7 held-out target benchmarks. Trained on pooled 13 benchmarks. **Bold**: best; *italic*: second best. ↓/↑: lower/higher is better.

Calibration Method	Transfer Protocol	Macro ECE (%) ↓	Macro Ada-ECE (%) ↓	Macro Brier ↓	Macro AUROC ↑
Naive Confidence (NC)	Zero-Shot Base	19.40	19.39	0.2356	<i>0.765</i>
Naive Confidence (NC)	Target Adapted (Saerens-EM)	37.57	37.57	0.3669	0.713
Temperature Scaling (TS)	Zero-Shot Base	15.41	21.32	0.2363	<i>0.765</i>
Temperature Scaling (TS)	Target Adapted (Saerens-EM)	40.61	40.61	0.4061	<i>0.765</i>
Adaptive TS (ATS)	Zero-Shot Base	<i>15.44</i>	21.72	<i>0.2362</i>	0.767
Adaptive TS (ATS)	Target Adapted (Saerens-EM)	40.61	40.61	0.4061	0.767
Platt Scaling (1D)	Zero-Shot Base	24.43	25.41	0.2800	<i>0.765</i>
Platt Scaling (1D)	Target Adapted (Saerens-EM)	56.23	56.23	0.5623	<i>0.765</i>
Trajectory Platt (5D)	Zero-Shot Base	19.83	<i>20.03</i>	0.2467	0.723
Trajectory Platt (5D)	Target Adapted (Saerens-EM)	41.67	41.80	0.4131	0.723

Table 15: **Leave-One-Dataset-Out (LODO) Cross-Domain Transfer (MQT-LLaVA 7B)**. Macro-averaged across Core 7 held-out target benchmarks. Trained on pooled 13 benchmarks. **Bold**: best; *italic*: second best. ↓/↑: lower/higher is better.

Calibration Method	Transfer Protocol	Macro ECE (%) ↓	Macro Ada-ECE (%) ↓	Macro Brier ↓	Macro AUROC ↑
Naive Confidence (NC)	Zero-Shot Base	24.48	24.48	0.2542	<i>0.753</i>
Naive Confidence (NC)	Target Adapted (Saerens-EM)	46.64	46.64	0.4495	0.673
Temperature Scaling (TS)	Zero-Shot Base	19.50	19.83	0.2247	<i>0.753</i>
Temperature Scaling (TS)	Target Adapted (Saerens-EM)	49.56	49.56	0.4956	<i>0.753</i>
Adaptive TS (ATS)	Zero-Shot Base	19.26	19.62	0.2233	0.756
Adaptive TS (ATS)	Target Adapted (Saerens-EM)	49.56	49.56	0.4956	0.756
Platt Scaling (1D)	Zero-Shot Base	<i>14.80</i>	<i>14.82</i>	<i>0.2114</i>	<i>0.753</i>
Platt Scaling (1D)	Target Adapted (Saerens-EM)	24.14	24.08	0.2617	<i>0.753</i>
Trajectory Platt (5D)	Zero-Shot Base	12.33	12.26	0.2070	0.734
Trajectory Platt (5D)	Target Adapted (Saerens-EM)	21.52	21.57	0.2567	0.734

1.10 Group 4: Leave-One-Dataset-Out (LODO) Cross-Domain Transfer

2 Group 3: Temperature Transfer Robustness Ablation

Evaluating calibrators trained at $T = 0.0$ and tested across $T \in \{0.0, 0.3, 0.6, 1.0, 1.5\}$ and temperature Mean.

2.1 Temperature Transfer: Macro-Averaged Across 4 Benchmarks

Table 16: **Temperature Transfer Robustness (Macro-Averaged Across 4 Benchmarks) (M3-LLaVA 7B)**. Evaluated across sampling temperatures $T \in \{0.0, 0.3, 0.6, 1.0, 1.5\}$ and Mean (trained at $T = 0.0$). **Bold**: best; *italic*: second best. $\downarrow/\uparrow$: lower/higher is better.

Calibration Method	ECE (%) $\downarrow$	Ada-ECE (%) $\downarrow$	Brier $\downarrow$	AUROC $\uparrow$
Sampling Temperature $T = 0.0$				
Naive Confidence (NC)	13.02	13.34	0.1759	0.794
Temperature Scaling (TS)	10.71	10.82	0.1646	0.794
Adaptive TS (ATS)	10.61	10.86	<i>0.1641</i>	<i>0.797</i>
Platt Scaling (1D)	<i>6.02</i>	5.83	0.1515	0.794
Trajectory Platt (5D)	5.54	<i>6.17</i>	0.1515	0.798
Sampling Temperature $T = 0.3$				
Naive Confidence (NC)	28.46	27.71	0.2767	0.780
Temperature Scaling (TS)	25.61	24.92	0.2486	0.780
Adaptive TS (ATS)	25.50	24.92	0.2486	<i>0.782</i>
Platt Scaling (1D)	<i>23.60</i>	<i>22.85</i>	<i>0.2359</i>	0.780
Trajectory Platt (5D)	22.47	21.83	0.2308	0.784
Sampling Temperature $T = 0.6$				
Naive Confidence (NC)	18.97	18.02	0.2045	0.799
Temperature Scaling (TS)	14.84	14.49	0.1815	0.799
Adaptive TS (ATS)	14.77	14.80	0.1811	<i>0.802</i>
Platt Scaling (1D)	<i>11.45</i>	<i>11.47</i>	<i>0.1693</i>	0.799
Trajectory Platt (5D)	11.07	10.51	0.1668	0.803
Sampling Temperature $T = 1.0$				
Naive Confidence (NC)	11.89	11.75	0.1582	<i>0.840</i>
Temperature Scaling (TS)	9.54	9.41	0.1458	<i>0.840</i>
Adaptive TS (ATS)	9.78	9.71	0.1452	0.841
Platt Scaling (1D)	5.78	5.20	0.1351	<i>0.840</i>
Trajectory Platt (5D)	<i>7.67</i>	<i>7.01</i>	<i>0.1371</i>	<i>0.840</i>
Sampling Temperature $T = 1.5$				
Naive Confidence (NC)	9.92	10.03	0.1426	<i>0.841</i>
Temperature Scaling (TS)	<i>9.25</i>	9.07	<i>0.1390</i>	<i>0.841</i>
Adaptive TS (ATS)	9.30	9.41	0.1386	0.843
Platt Scaling (1D)	9.07	<i>9.12</i>	<i>0.1390</i>	<i>0.841</i>
Trajectory Platt (5D)	10.92	10.68	0.1464	0.840
Mean (Averaged Across Temperatures)				
Naive Confidence (NC)	16.45	16.17	0.1916	<i>0.811</i>
Temperature Scaling (TS)	13.99	13.74	0.1759	<i>0.811</i>
Adaptive TS (ATS)	13.99	13.94	0.1755	0.813
Platt Scaling (1D)	11.19	10.89	0.1662	<i>0.811</i>
Trajectory Platt (5D)	<i>11.53</i>	<i>11.24</i>	<i>0.1665</i>	0.813

Table 17: **Temperature Transfer Robustness (Macro-Averaged Across 4 Benchmarks) (MQT-LLaVA 7B)**. Evaluated across sampling temperatures $T \in \{0.0, 0.3, 0.6, 1.0, 1.5\}$ and Mean (trained at $T = 0.0$). **Bold**: best; *italic*: second best. $\downarrow/\uparrow$: lower/higher is better.

Calibration Method	ECE (%) $\downarrow$	Ada-ECE (%) $\downarrow$	Brier $\downarrow$	AUROC $\uparrow$
Sampling Temperature $T = 0.0$				
Naive Confidence (NC)	22.65	22.51	0.2301	0.779
Temperature Scaling (TS)	15.11	15.72	0.1927	0.779
Adaptive TS (ATS)	14.99	15.44	0.1920	<i>0.780</i>
Platt Scaling (1D)	<i>6.79</i>	<i>7.41</i>	<i>0.1598</i>	0.779
Trajectory Platt (5D)	5.32	6.53	0.1532	0.789
Sampling Temperature $T = 0.3$				
Naive Confidence (NC)	41.12	40.92	0.3853	0.758
Temperature Scaling (TS)	28.26	28.20	0.2669	0.758
Adaptive TS (ATS)	28.73	28.75	0.2682	<i>0.762</i>
Platt Scaling (1D)	<i>24.89</i>	<i>24.42</i>	<i>0.2500</i>	0.758
Trajectory Platt (5D)	20.66	20.61	0.2198	0.786
Sampling Temperature $T = 0.6$				
Naive Confidence (NC)	34.52	34.17	0.3170	0.763
Temperature Scaling (TS)	21.79	21.89	0.2245	0.763
Adaptive TS (ATS)	21.86	21.95	0.2250	<i>0.765</i>
Platt Scaling (1D)	<i>13.50</i>	<i>13.28</i>	<i>0.1855</i>	0.763
Trajectory Platt (5D)	10.48	10.51	0.1633	0.794
Sampling Temperature $T = 1.0$				
Naive Confidence (NC)	25.72	25.83	0.2443	0.789
Temperature Scaling (TS)	17.81	18.31	0.1988	0.789
Adaptive TS (ATS)	17.49	18.21	0.1979	<i>0.791</i>
Platt Scaling (1D)	<i>7.14</i>	<i>7.08</i>	<i>0.1529</i>	0.789
Trajectory Platt (5D)	5.03	5.13	0.1393	0.817
Sampling Temperature $T = 1.5$				
Naive Confidence (NC)	20.45	20.69	0.2017	0.805
Temperature Scaling (TS)	19.86	19.93	0.1983	0.805
Adaptive TS (ATS)	19.52	19.88	0.1962	<i>0.806</i>
Platt Scaling (1D)	<i>8.72</i>	8.61	<i>0.1507</i>	0.805
Trajectory Platt (5D)	7.79	<i>8.69</i>	0.1439	0.820
Mean (Averaged Across Temperatures)				
Naive Confidence (NC)	28.89	28.82	0.2757	0.779
Temperature Scaling (TS)	20.57	20.81	0.2162	0.779
Adaptive TS (ATS)	20.52	20.85	0.2159	<i>0.781</i>
Platt Scaling (1D)	<i>12.21</i>	<i>12.16</i>	<i>0.1798</i>	0.779
Trajectory Platt (5D)	9.86	10.29	0.1639	0.801

2.2 Temperature Transfer: POPE

Table 18: **Temperature Transfer Robustness on pope (M3-LLaVA 7B)**. Evaluated across sampling temperatures $T \in \{0.0, 0.3, 0.6, 1.0, 1.5\}$ and Mean (trained at $T = 0.0$). **Bold**: best; *italic*: second best. $\downarrow/\uparrow$: lower/higher is better.

Calibration Method	ECE (%) $\downarrow$	Ada-ECE (%) $\downarrow$	Brier $\downarrow$	AUROC $\uparrow$
Sampling Temperature $T = 0.0$				
Naive Confidence (NC)	3.45	4.40	<i>0.0882</i>	0.739
Temperature Scaling (TS)	4.27	4.37	0.0883	0.739
Adaptive TS (ATS)	4.42	4.27	0.0879	<i>0.750</i>
Platt Scaling (1D)	<i>4.19</i>	<i>4.32</i>	<i>0.0882</i>	0.739
Trajectory Platt (5D)	4.82	6.62	0.0887	0.751
Sampling Temperature $T = 0.3$				
Naive Confidence (NC)	9.83	8.69	0.0980	0.764
Temperature Scaling (TS)	<i>9.69</i>	<i>8.53</i>	<i>0.0972</i>	0.764
Adaptive TS (ATS)	9.66	8.54	0.0968	0.772
Platt Scaling (1D)	<i>9.69</i>	8.55	<i>0.0972</i>	0.764
Trajectory Platt (5D)	9.71	8.43	0.0975	<i>0.770</i>
Sampling Temperature $T = 0.6$				
Naive Confidence (NC)	7.00	6.53	0.0912	0.786
Temperature Scaling (TS)	6.73	6.02	0.0904	0.786
Adaptive TS (ATS)	6.40	6.42	0.0901	0.796
Platt Scaling (1D)	<i>6.64</i>	<i>5.97</i>	0.0904	0.786
Trajectory Platt (5D)	7.45	5.76	<i>0.0902</i>	<i>0.794</i>
Sampling Temperature $T = 1.0$				
Naive Confidence (NC)	4.34	4.56	<i>0.0882</i>	0.826
Temperature Scaling (TS)	<i>5.08</i>	<i>4.43</i>	0.0883	0.826
Adaptive TS (ATS)	5.15	4.38	<i>0.0882</i>	<i>0.828</i>
Platt Scaling (1D)	5.16	4.47	0.0883	0.826
Trajectory Platt (5D)	6.77	4.82	0.0872	0.832
Sampling Temperature $T = 1.5$				
Naive Confidence (NC)	5.38	6.14	0.0953	0.876
Temperature Scaling (TS)	<i>6.83</i>	<i>6.80</i>	<i>0.0970</i>	0.876
Adaptive TS (ATS)	7.34	7.98	0.0977	0.880
Platt Scaling (1D)	7.25	6.81	0.0972	0.876
Trajectory Platt (5D)	9.71	9.39	0.1004	<i>0.879</i>
Mean (Averaged Across Temperatures)				
Naive Confidence (NC)	6.00	6.06	<i>0.0922</i>	<i>0.798</i>
Temperature Scaling (TS)	<i>6.52</i>	<i>6.03</i>	<i>0.0922</i>	<i>0.798</i>
Adaptive TS (ATS)	6.59	6.32	0.0921	0.805
Platt Scaling (1D)	6.59	6.02	0.0923	<i>0.798</i>
Trajectory Platt (5D)	7.69	7.00	0.0928	0.805

Table 19: **Temperature Transfer Robustness on pope (MQT-LLaVA 7B)**. Evaluated across sampling temperatures $T \in \{0.0, 0.3, 0.6, 1.0, 1.5\}$ and Mean (trained at $T = 0.0$). **Bold**: best; *italic*: second best. $\downarrow/\uparrow$: lower/higher is better.

Calibration Method	ECE (%) $\downarrow$	Ada-ECE (%) $\downarrow$	Brier $\downarrow$	AUROC $\uparrow$
Sampling Temperature $T = 0.0$				
Naive Confidence (NC)	4.01	4.30	0.0903	0.876
Temperature Scaling (TS)	3.52	4.93	<i>0.0902</i>	0.876
Adaptive TS (ATS)	3.75	4.90	<i>0.0902</i>	<i>0.874</i>
Platt Scaling (1D)	3.97	4.64	0.0893	0.876
Trajectory Platt (5D)	<i>3.55</i>	<i>4.48</i>	0.0903	0.873
Sampling Temperature $T = 0.3$				
Naive Confidence (NC)	10.43	10.43	0.1108	0.872
Temperature Scaling (TS)	<i>9.42</i>	<i>9.18</i>	<i>0.1043</i>	0.872
Adaptive TS (ATS)	9.47	9.26	0.1048	<i>0.871</i>
Platt Scaling (1D)	9.20	8.91	0.1031	0.872
Trajectory Platt (5D)	10.26	10.17	0.1108	0.868
Sampling Temperature $T = 0.6$				
Naive Confidence (NC)	8.53	8.26	0.1098	0.852
Temperature Scaling (TS)	<i>6.12</i>	6.16	<i>0.1036</i>	0.852
Adaptive TS (ATS)	6.10	<i>6.15</i>	0.1038	<i>0.851</i>
Platt Scaling (1D)	6.78	5.66	0.1026	0.852
Trajectory Platt (5D)	7.67	6.88	0.1068	0.846
Sampling Temperature $T = 1.0$				
Naive Confidence (NC)	5.39	5.84	0.1081	0.855
Temperature Scaling (TS)	4.31	4.94	<i>0.1046</i>	0.855
Adaptive TS (ATS)	<i>3.97</i>	5.59	<i>0.1046</i>	0.855
Platt Scaling (1D)	4.42	<i>4.83</i>	0.1037	0.855
Trajectory Platt (5D)	2.34	3.68	0.1048	<i>0.852</i>
Sampling Temperature $T = 1.5$				
Naive Confidence (NC)	8.46	9.36	0.1545	0.810
Temperature Scaling (TS)	8.73	7.36	0.1497	0.810
Adaptive TS (ATS)	8.47	7.05	0.1496	0.810
Platt Scaling (1D)	<i>6.48</i>	<i>7.12</i>	0.1473	0.810
Trajectory Platt (5D)	6.20	7.43	<i>0.1487</i>	<i>0.805</i>
Mean (Averaged Across Temperatures)				
Naive Confidence (NC)	7.36	7.64	0.1147	0.853
Temperature Scaling (TS)	6.42	<i>6.52</i>	<i>0.1105</i>	0.853
Adaptive TS (ATS)	6.35	6.59	0.1106	<i>0.852</i>
Platt Scaling (1D)	<i>6.17</i>	6.23	0.1092	0.853
Trajectory Platt (5D)	6.01	6.53	0.1123	0.849

2.3 Temperature Transfer: ScienceQA

Table 20: **Temperature Transfer Robustness on scienceqa (M3-LLaVA 7B)**. Evaluated across sampling temperatures $T \in \{0.0, 0.3, 0.6, 1.0, 1.5\}$ and Mean (trained at $T = 0.0$). **Bold**: best; *italic*: second best. $\downarrow/\uparrow$: lower/higher is better.

Calibration Method	ECE (%) $\downarrow$	Ada-ECE (%) $\downarrow$	Brier $\downarrow$	AUROC $\uparrow$
Sampling Temperature $T = 0.0$				
Naive Confidence (NC)	22.16	21.93	0.2368	0.807
Temperature Scaling (TS)	13.10	12.68	0.1924	0.807
Adaptive TS (ATS)	12.98	13.00	0.1918	<i>0.808</i>
Platt Scaling (1D)	4.80	4.99	<i>0.1732</i>	0.807
Trajectory Platt (5D)	<i>5.17</i>	<i>6.15</i>	0.1719	0.813
Sampling Temperature $T = 0.3$				
Naive Confidence (NC)	32.97	32.97	0.3341	0.786
Temperature Scaling (TS)	<i>25.55</i>	25.61	<i>0.2658</i>	0.786
Adaptive TS (ATS)	25.41	25.48	0.2640	<i>0.792</i>
Platt Scaling (1D)	25.58	<i>25.08</i>	0.2660	0.786
Trajectory Platt (5D)	25.62	24.85	<i>0.2658</i>	0.793
Sampling Temperature $T = 0.6$				
Naive Confidence (NC)	29.33	29.02	0.2947	<i>0.797</i>
Temperature Scaling (TS)	18.72	18.70	0.2211	<i>0.797</i>
Adaptive TS (ATS)	18.60	18.93	0.2199	0.799
Platt Scaling (1D)	<i>16.17</i>	<i>16.77</i>	0.2082	<i>0.797</i>
Trajectory Platt (5D)	16.02	16.24	<i>0.2093</i>	0.799
Sampling Temperature $T = 1.0$				
Naive Confidence (NC)	24.79	24.47	0.2492	<i>0.817</i>
Temperature Scaling (TS)	15.07	14.82	0.1985	<i>0.817</i>
Adaptive TS (ATS)	14.81	15.88	0.1971	0.820
Platt Scaling (1D)	6.05	5.67	0.1726	<i>0.817</i>
Trajectory Platt (5D)	<i>8.14</i>	<i>7.44</i>	<i>0.1757</i>	0.811
Sampling Temperature $T = 1.5$				
Naive Confidence (NC)	18.84	19.07	0.2272	<i>0.789</i>
Temperature Scaling (TS)	13.83	13.84	0.2068	<i>0.789</i>
Adaptive TS (ATS)	14.18	14.76	0.2052	0.795
Platt Scaling (1D)	<i>9.20</i>	<i>8.90</i>	0.1905	<i>0.789</i>
Trajectory Platt (5D)	8.54	7.91	<i>0.1919</i>	0.783
Mean (Averaged Across Temperatures)				
Naive Confidence (NC)	25.62	25.49	0.2684	0.799
Temperature Scaling (TS)	17.25	17.13	0.2169	0.799
Adaptive TS (ATS)	17.20	17.61	0.2156	0.803
Platt Scaling (1D)	12.36	12.28	0.2021	0.799
Trajectory Platt (5D)	<i>12.70</i>	<i>12.52</i>	<i>0.2029</i>	<i>0.800</i>

Table 21: **Temperature Transfer Robustness on scienceqa (MQT-LLaVA 7B)**. Evaluated across sampling temperatures $T \in \{0.0, 0.3, 0.6, 1.0, 1.5\}$ and Mean (trained at $T = 0.0$). **Bold**: best; *italic*: second best. $\downarrow/\uparrow$: lower/higher is better.

Calibration Method	ECE (%) $\downarrow$	Ada-ECE (%) $\downarrow$	Brier $\downarrow$	AUROC $\uparrow$
Sampling Temperature $T = 0.0$				
Naive Confidence (NC)	23.65	23.17	0.2579	0.760
Temperature Scaling (TS)	9.32	10.23	0.1975	0.760
Adaptive TS (ATS)	8.73	9.38	0.1966	<i>0.761</i>
Platt Scaling (1D)	<i>7.99</i>	<i>6.91</i>	<i>0.1916</i>	0.760
Trajectory Platt (5D)	6.46	5.72	0.1860	0.769
Sampling Temperature $T = 0.3$				
Naive Confidence (NC)	34.72	33.93	0.3405	<i>0.740</i>
Temperature Scaling (TS)	23.17	23.17	0.2567	<i>0.740</i>
Adaptive TS (ATS)	22.78	<i>22.78</i>	0.2540	0.731
Platt Scaling (1D)	23.39	22.36	<i>0.2554</i>	<i>0.740</i>
Trajectory Platt (5D)	<i>23.04</i>	22.95	0.2575	0.756
Sampling Temperature $T = 0.6$				
Naive Confidence (NC)	30.38	29.26	0.3037	0.757
Temperature Scaling (TS)	14.69	15.04	0.2163	0.757
Adaptive TS (ATS)	14.41	14.73	0.2141	<i>0.759</i>
Platt Scaling (1D)	<i>13.17</i>	12.45	<i>0.2105</i>	0.757
Trajectory Platt (5D)	11.66	<i>12.50</i>	0.2056	0.768
Sampling Temperature $T = 1.0$				
Naive Confidence (NC)	24.42	24.42	0.2651	0.767
Temperature Scaling (TS)	10.57	11.96	0.2000	0.767
Adaptive TS (ATS)	9.72	10.41	0.1978	<i>0.770</i>
Platt Scaling (1D)	<i>7.07</i>	<i>6.82</i>	<i>0.1902</i>	0.767
Trajectory Platt (5D)	6.28	5.31	0.1838	0.777
Sampling Temperature $T = 1.5$				
Naive Confidence (NC)	17.63	17.71	0.2298	0.767
Temperature Scaling (TS)	<i>10.88</i>	12.64	<i>0.2020</i>	0.767
Adaptive TS (ATS)	10.66	13.18	0.2005	<i>0.770</i>
Platt Scaling (1D)	12.93	12.97	0.2062	0.767
Trajectory Platt (5D)	12.83	<i>12.86</i>	0.2033	0.775
Mean (Averaged Across Temperatures)				
Naive Confidence (NC)	26.16	25.70	0.2794	<i>0.758</i>
Temperature Scaling (TS)	13.73	14.61	0.2145	<i>0.758</i>
Adaptive TS (ATS)	13.26	14.09	0.2126	<i>0.758</i>
Platt Scaling (1D)	<i>12.91</i>	<i>12.30</i>	<i>0.2108</i>	<i>0.758</i>
Trajectory Platt (5D)	12.05	11.87	0.2072	0.769

2.4 Temperature Transfer: TextVQA

Table 22: **Temperature Transfer Robustness on textvqa (M3-LLaVA 7B)**. Evaluated across sampling temperatures $T \in \{0.0, 0.3, 0.6, 1.0, 1.5\}$ and Mean (trained at $T = 0.0$). **Bold**: best; *italic*: second best. $\downarrow/\uparrow$: lower/higher is better.

Calibration Method	ECE (%) $\downarrow$	Ada-ECE (%) $\downarrow$	Brier $\downarrow$	AUROC $\uparrow$
Sampling Temperature $T = 0.0$				
Naive Confidence (NC)	10.50	11.07	0.1714	0.839
Temperature Scaling (TS)	10.38	11.14	0.1712	0.839
Adaptive TS (ATS)	10.15	11.31	0.1710	0.839
Platt Scaling (1D)	<i>8.47</i>	<i>8.22</i>	0.1615	0.839
Trajectory Platt (5D)	5.78	5.35	<i>0.1628</i>	<i>0.835</i>
Sampling Temperature $T = 0.3$				
Naive Confidence (NC)	27.03	25.61	0.2691	0.804
Temperature Scaling (TS)	26.09	24.82	0.2611	0.804
Adaptive TS (ATS)	25.61	24.61	0.2584	<i>0.805</i>
Platt Scaling (1D)	<i>24.52</i>	<i>22.97</i>	<i>0.2510</i>	0.804
Trajectory Platt (5D)	20.29	20.14	0.2314	0.808
Sampling Temperature $T = 0.6$				
Naive Confidence (NC)	15.85	14.07	0.1849	<i>0.846</i>
Temperature Scaling (TS)	13.76	13.27	0.1815	<i>0.846</i>
Adaptive TS (ATS)	14.12	13.21	0.1802	<i>0.846</i>
Platt Scaling (1D)	<i>11.19</i>	<i>10.87</i>	<i>0.1697</i>	<i>0.846</i>
Trajectory Platt (5D)	7.48	6.67	0.1575	0.852
Sampling Temperature $T = 1.0$				
Naive Confidence (NC)	<i>6.85</i>	<i>6.87</i>	0.1256	<i>0.902</i>
Temperature Scaling (TS)	7.05	7.79	0.1263	<i>0.902</i>
Adaptive TS (ATS)	8.55	8.43	0.1272	0.901
Platt Scaling (1D)	4.51	4.61	0.1207	<i>0.902</i>
Trajectory Platt (5D)	7.96	7.61	<i>0.1254</i>	0.904
Sampling Temperature $T = 1.5$				
Naive Confidence (NC)	<i>8.51</i>	8.63	0.1221	0.871
Temperature Scaling (TS)	8.26	8.91	<i>0.1233</i>	0.871
Adaptive TS (ATS)	8.76	<i>8.73</i>	0.1244	0.871
Platt Scaling (1D)	10.44	11.32	0.1313	0.871
Trajectory Platt (5D)	15.83	15.77	0.1555	0.871
Mean (Averaged Across Temperatures)				
Naive Confidence (NC)	13.75	13.25	0.1746	<i>0.852</i>
Temperature Scaling (TS)	13.11	13.19	0.1727	<i>0.852</i>
Adaptive TS (ATS)	13.44	13.26	0.1722	<i>0.852</i>
Platt Scaling (1D)	<i>11.83</i>	<i>11.60</i>	<i>0.1669</i>	<i>0.852</i>
Trajectory Platt (5D)	11.47	11.11	0.1665	0.854

Table 23: **Temperature Transfer Robustness on textvqa (MQT-LLaVA 7B)**. Evaluated across sampling temperatures $T \in \{0.0, 0.3, 0.6, 1.0, 1.5\}$ and Mean (trained at $T = 0.0$). **Bold**: best; *italic*: second best. $\downarrow/\uparrow$: lower/higher is better.

Calibration Method	ECE (%) $\downarrow$	Ada-ECE (%) $\downarrow$	Brier $\downarrow$	AUROC $\uparrow$
Sampling Temperature $T = 0.0$				
Naive Confidence (NC)	32.27	31.91	0.3286	0.684
Temperature Scaling (TS)	18.21	18.32	0.2503	0.684
Adaptive TS (ATS)	18.36	18.36	0.2500	<i>0.689</i>
Platt Scaling (1D)	<i>8.95</i>	<i>10.46</i>	<i>0.2148</i>	0.684
Trajectory Platt (5D)	5.25	7.76	0.1948	0.719
Sampling Temperature $T = 0.3$				
Naive Confidence (NC)	56.45	56.45	0.5294	0.663
Temperature Scaling (TS)	<i>30.04</i>	30.04	0.3007	0.663
Adaptive TS (ATS)	31.55	31.85	<i>0.2997</i>	<i>0.698</i>
Platt Scaling (1D)	30.12	<i>29.53</i>	0.3187	0.663
Trajectory Platt (5D)	16.54	16.63	0.2147	0.761
Sampling Temperature $T = 0.6$				
Naive Confidence (NC)	48.08	48.08	0.4405	0.674
Temperature Scaling (TS)	25.32	25.32	0.2724	0.674
Adaptive TS (ATS)	25.49	25.49	0.2714	<i>0.683</i>
Platt Scaling (1D)	<i>16.80</i>	<i>17.76</i>	<i>0.2479</i>	0.674
Trajectory Platt (5D)	9.06	8.16	0.1763	0.774
Sampling Temperature $T = 1.0$				
Naive Confidence (NC)	37.38	37.38	0.3509	0.698
Temperature Scaling (TS)	22.65	22.65	0.2534	0.698
Adaptive TS (ATS)	22.73	23.30	0.2527	<i>0.706</i>
Platt Scaling (1D)	<i>10.82</i>	<i>10.61</i>	<i>0.1997</i>	0.698
Trajectory Platt (5D)	4.78	5.47	0.1538	0.802
Sampling Temperature $T = 1.5$				
Naive Confidence (NC)	31.24	31.24	0.2717	0.759
Temperature Scaling (TS)	28.01	28.01	0.2442	0.759
Adaptive TS (ATS)	28.06	28.48	0.2442	<i>0.762</i>
Platt Scaling (1D)	<i>8.41</i>	<i>7.92</i>	<i>0.1578</i>	0.759
Trajectory Platt (5D)	5.66	6.85	0.1333	0.810
Mean (Averaged Across Temperatures)				
Naive Confidence (NC)	41.08	41.01	0.3842	0.696
Temperature Scaling (TS)	24.84	24.87	0.2642	0.696
Adaptive TS (ATS)	25.24	25.50	0.2636	<i>0.708</i>
Platt Scaling (1D)	<i>15.02</i>	<i>15.26</i>	<i>0.2278</i>	0.696
Trajectory Platt (5D)	8.26	8.97	0.1746	0.773

2.5 Temperature Transfer: VizWiz-VQA

Table 24: **Temperature Transfer Robustness on vizwiz-vqa (M3-LLaVA 7B)**. Evaluated across sampling temperatures $T \in \{0.0, 0.3, 0.6, 1.0, 1.5\}$ and Mean (trained at $T = 0.0$). **Bold**: best; *italic*: second best. $\downarrow/\uparrow$: lower/higher is better.

Calibration Method	ECE (%) $\downarrow$	Ada-ECE (%) $\downarrow$	Brier $\downarrow$	AUROC $\uparrow$
Sampling Temperature $T = 0.0$				
Naive Confidence (NC)	15.97	15.97	0.2071	<i>0.793</i>
Temperature Scaling (TS)	15.09	15.09	0.2065	<i>0.793</i>
Adaptive TS (ATS)	14.87	14.87	<i>0.2058</i>	<i>0.793</i>
Platt Scaling (1D)	<i>6.64</i>	5.80	0.1828	<i>0.793</i>
Trajectory Platt (5D)	6.38	<i>6.57</i>	0.1828	0.794
Sampling Temperature $T = 0.3$				
Naive Confidence (NC)	44.03	43.59	0.4055	0.764
Temperature Scaling (TS)	41.10	40.70	0.3701	0.764
Adaptive TS (ATS)	41.33	41.05	0.3751	0.762
Platt Scaling (1D)	<i>34.62</i>	<i>34.81</i>	<i>0.3295</i>	0.764
Trajectory Platt (5D)	34.27	33.91	0.3284	<i>0.763</i>
Sampling Temperature $T = 0.6$				
Naive Confidence (NC)	23.72	22.45	0.2470	0.768
Temperature Scaling (TS)	20.17	19.98	0.2331	0.768
Adaptive TS (ATS)	19.98	20.62	0.2340	0.768
Platt Scaling (1D)	11.80	12.25	0.2090	0.768
Trajectory Platt (5D)	<i>13.33</i>	<i>13.37</i>	<i>0.2103</i>	<i>0.766</i>
Sampling Temperature $T = 1.0$				
Naive Confidence (NC)	11.58	11.10	0.1699	<i>0.815</i>
Temperature Scaling (TS)	10.95	10.61	0.1701	<i>0.815</i>
Adaptive TS (ATS)	10.61	10.17	0.1683	0.816
Platt Scaling (1D)	7.40	6.03	0.1588	<i>0.815</i>
Trajectory Platt (5D)	<i>7.80</i>	<i>8.19</i>	<i>0.1600</i>	0.812
Sampling Temperature $T = 1.5$				
Naive Confidence (NC)	<i>6.96</i>	<i>6.27</i>	0.1259	0.828
Temperature Scaling (TS)	8.10	6.71	0.1290	0.828
Adaptive TS (ATS)	6.90	6.17	<i>0.1271</i>	0.828
Platt Scaling (1D)	9.40	9.46	0.1368	0.828
Trajectory Platt (5D)	9.60	9.67	0.1379	<i>0.826</i>
Mean (Averaged Across Temperatures)				
Naive Confidence (NC)	20.45	19.88	0.2311	0.793
Temperature Scaling (TS)	19.08	18.62	0.2218	0.793
Adaptive TS (ATS)	18.74	18.58	0.2221	0.793
Platt Scaling (1D)	13.97	13.67	0.2034	0.793
Trajectory Platt (5D)	<i>14.28</i>	<i>14.34</i>	<i>0.2039</i>	<i>0.792</i>

Table 25: **Temperature Transfer Robustness on vizwiz-vqa (MQT-LLaVA 7B)**. Evaluated across sampling temperatures $T \in \{0.0, 0.3, 0.6, 1.0, 1.5\}$ and Mean (trained at $T = 0.0$). **Bold**: best; *italic*: second best. $\downarrow/\uparrow$: lower/higher is better.

Calibration Method	ECE (%) $\downarrow$	Ada-ECE (%) $\downarrow$	Brier $\downarrow$	AUROC $\uparrow$
Sampling Temperature $T = 0.0$				
Naive Confidence (NC)	30.66	30.66	0.2435	0.797
Temperature Scaling (TS)	29.41	29.41	0.2328	0.797
Adaptive TS (ATS)	29.11	29.11	0.2312	<i>0.796</i>
Platt Scaling (1D)	<i>6.27</i>	7.62	<i>0.1433</i>	0.797
Trajectory Platt (5D)	6.03	<i>8.17</i>	0.1418	0.795
Sampling Temperature $T = 0.3$				
Naive Confidence (NC)	62.87	62.87	0.5606	<i>0.756</i>
Temperature Scaling (TS)	50.41	50.41	0.4059	<i>0.756</i>
Adaptive TS (ATS)	51.12	51.12	0.4144	0.750
Platt Scaling (1D)	<i>36.87</i>	<i>36.87</i>	<i>0.3228</i>	<i>0.756</i>
Trajectory Platt (5D)	32.79	32.71	0.2960	0.759
Sampling Temperature $T = 0.6$				
Naive Confidence (NC)	51.06	51.06	0.4138	<i>0.770</i>
Temperature Scaling (TS)	41.04	41.04	0.3059	<i>0.770</i>
Adaptive TS (ATS)	41.42	41.42	0.3106	0.765
Platt Scaling (1D)	<i>17.23</i>	<i>17.23</i>	<i>0.1811</i>	<i>0.770</i>
Trajectory Platt (5D)	13.53	14.51	0.1646	0.789
Sampling Temperature $T = 1.0$				
Naive Confidence (NC)	35.69	35.69	0.2530	0.837
Temperature Scaling (TS)	33.71	33.71	0.2374	0.837
Adaptive TS (ATS)	33.54	33.54	0.2365	<i>0.834</i>
Platt Scaling (1D)	6.25	<i>6.08</i>	<i>0.1182</i>	0.837
Trajectory Platt (5D)	<i>6.71</i>	6.06	0.1148	0.837
Sampling Temperature $T = 1.5$				
Naive Confidence (NC)	24.45	24.45	0.1509	<i>0.884</i>
Temperature Scaling (TS)	31.84	31.70	0.1971	<i>0.884</i>
Adaptive TS (ATS)	30.90	30.81	0.1904	0.882
Platt Scaling (1D)	<i>7.08</i>	6.42	<i>0.0916</i>	<i>0.884</i>
Trajectory Platt (5D)	6.46	<i>7.61</i>	0.0905	0.890
Mean (Averaged Across Temperatures)				
Naive Confidence (NC)	40.95	40.95	0.3244	<i>0.809</i>
Temperature Scaling (TS)	37.28	37.25	0.2758	<i>0.809</i>
Adaptive TS (ATS)	37.22	37.20	0.2766	0.805
Platt Scaling (1D)	<i>14.74</i>	<i>14.84</i>	<i>0.1714</i>	<i>0.809</i>
Trajectory Platt (5D)	13.11	13.81	0.1615	0.814

Table 26: **Calibration Benchmark on ai2d (M3-LLaVA 7B)**. **Bold**: best; *italic*: second best. ↓/↑: lower/higher is better.

Calibration Method	Regime / Sampling	ECE (%) ↓	Ada-ECE (%) ↓	Brier ↓	AUROC ↑
Naive Confidence (NC)	Single-Pass ($T = 0.0$)	42.00	42.00	0.4200	0.614
Temperature Scaling (TS)	Single-Pass ($T = 0.0$)	8.62	12.18	0.2477	0.614
Adaptive TS (ATS)	Single-Pass ($T = 0.0$)	8.64	13.15	0.2476	<i>0.625</i>
Platt Scaling (1D)	Single-Pass ($T = 0.0$)	5.63	7.70	<i>0.2370</i>	0.614
LN-Entropy	Multi-Pass ($T = 0.5, K = 10$)	-	-	-	-
Semantic Entropy	Multi-Pass ($T = 0.5, K = 10$)	-	-	-	-
EigenScore	Multi-Pass ($T = 0.5, K = 10$)	-	-	-	-
UMPIRE	Multi-Pass ($T = 0.5, K = 10$)	-	-	-	-
Trajectory Platt (5D)	Single-Pass ($T = 0.0$)	<i>8.61</i>	<i>11.15</i>	0.2289	0.661

Table 27: **Calibration Benchmark on ai2d (MQT-LLaVA 7B)**. **Bold**: best; *italic*: second best. ↓/↑: lower/higher is better.

Calibration Method	Regime / Sampling	ECE (%) ↓	Ada-ECE (%) ↓	Brier ↓	AUROC ↑
Naive Confidence (NC)	Single-Pass ($T = 0.0$)	27.15	26.92	0.2887	<i>0.746</i>
Temperature Scaling (TS)	Single-Pass ($T = 0.0$)	13.31	13.10	0.2197	<i>0.746</i>
Adaptive TS (ATS)	Single-Pass ($T = 0.0$)	13.38	14.08	0.2195	0.747
Platt Scaling (1D)	Single-Pass ($T = 0.0$)	5.92	<i>8.25</i>	0.2015	<i>0.746</i>
LN-Entropy	Multi-Pass ($T = 0.5, K = 10$)	-	-	-	-
Semantic Entropy	Multi-Pass ($T = 0.5, K = 10$)	-	-	-	-
EigenScore	Multi-Pass ($T = 0.5, K = 10$)	-	-	-	-
UMPIRE	Multi-Pass ($T = 0.5, K = 10$)	-	-	-	-
Trajectory Platt (5D)	Single-Pass ($T = 0.0$)	<i>6.27</i>	6.53	<i>0.2035</i>	0.739

3 Group 1: Per-Dataset In-Domain Calibration Benchmarks

Individual benchmark tables for all vision-language datasets under the 4-method, 4-metric schema.

3.1 Benchmark: AI2D

Table 28: **Calibration Benchmark on avqa (M3-LLaVA 7B)**. **Bold**: best; *italic*: second best. ↓/↑: lower/higher is better.

Calibration Method	Regime / Sampling	ECE (%) ↓	Ada-ECE (%) ↓	Brier ↓	AUROC ↑
Naive Confidence (NC)	Single-Pass ($T = 0.0$)	25.66	25.40	0.3194	0.599
Temperature Scaling (TS)	Single-Pass ($T = 0.0$)	<i>4.45</i>	17.96	0.2480	0.599
Adaptive TS (ATS)	Single-Pass ($T = 0.0$)	4.29	<i>16.94</i>	0.2477	<i>0.601</i>
Platt Scaling (1D)	Single-Pass ($T = 0.0$)	6.84	17.67	<i>0.2463</i>	0.599
LN-Entropy	Multi-Pass ($T = 0.5, K = 10$)	-	-	-	-
Semantic Entropy	Multi-Pass ($T = 0.5, K = 10$)	-	-	-	-
EigenScore	Multi-Pass ($T = 0.5, K = 10$)	-	-	-	-
UMPIRE	Multi-Pass ($T = 0.5, K = 10$)	-	-	-	-
Trajectory Platt (5D)	Single-Pass ($T = 0.0$)	5.26	7.45	0.2240	0.691

Table 29: **Calibration Benchmark on avqa (MQT-LLaVA 7B)**. **Bold**: best; *italic*: second best. ↓/↑: lower/higher is better.

Calibration Method	Regime / Sampling	ECE (%) ↓	Ada-ECE (%) ↓	Brier ↓	AUROC ↑
Naive Confidence (NC)	Single-Pass ($T = 0.0$)	27.88	28.00	0.3385	0.602
Temperature Scaling (TS)	Single-Pass ($T = 0.0$)	6.56	13.28	0.2491	0.602
Adaptive TS (ATS)	Single-Pass ($T = 0.0$)	6.62	<i>12.86</i>	0.2488	<i>0.604</i>
Platt Scaling (1D)	Single-Pass ($T = 0.0$)	1.18	13.66	<i>0.2449</i>	0.602
LN-Entropy	Multi-Pass ($T = 0.5, K = 10$)	-	-	-	-
Semantic Entropy	Multi-Pass ($T = 0.5, K = 10$)	-	-	-	-
EigenScore	Multi-Pass ($T = 0.5, K = 10$)	-	-	-	-
UMPIRE	Multi-Pass ($T = 0.5, K = 10$)	-	-	-	-
Trajectory Platt (5D)	Single-Pass ($T = 0.0$)	<i>5.02</i>	6.54	0.2332	0.654

3.2 Benchmark: AVQA

Table 30: **Calibration Benchmark on chartqa (M3-LLaVA 7B)**. **Bold**: best; *italic*: second best. $\downarrow/\uparrow$: lower/higher is better.

Calibration Method	Regime / Sampling	ECE (%) $\downarrow$	Ada-ECE (%) $\downarrow$	Brier $\downarrow$	AUROC $\uparrow$
Naive Confidence (NC)	Single-Pass ($T = 0.0$)	71.03	71.03	0.6151	<i>0.807</i>
Temperature Scaling (TS)	Single-Pass ($T = 0.0$)	44.47	44.47	0.2782	<i>0.807</i>
Adaptive TS (ATS)	Single-Pass ($T = 0.0$)	44.45	44.45	0.2781	<i>0.807</i>
Platt Scaling (1D)	Single-Pass ($T = 0.0$)	3.23	<i>3.62</i>	<i>0.0755</i>	<i>0.807</i>
Trajectory Platt (5D)	Single-Pass ($T = 0.0$)	<i>3.31</i>	2.88	0.0752	0.818

Table 31: **Calibration Benchmark on chartqa (MQT-LLaVA 7B)**. **Bold**: best; *italic*: second best. $\downarrow/\uparrow$: lower/higher is better.

Calibration Method	Regime / Sampling	ECE (%) $\downarrow$	Ada-ECE (%) $\downarrow$	Brier $\downarrow$	AUROC $\uparrow$
Naive Confidence (NC)	Single-Pass ($T = 0.0$)	71.30	71.30	0.6483	0.491
Temperature Scaling (TS)	Single-Pass ($T = 0.0$)	44.88	44.88	0.2779	0.491
Adaptive TS (ATS)	Single-Pass ($T = 0.0$)	44.87	44.87	0.2777	0.491
Platt Scaling (1D)	Single-Pass ($T = 0.0$)	<i>3.06</i>	<i>4.27</i>	0.0762	<i>0.509</i>
Trajectory Platt (5D)	Single-Pass ($T = 0.0$)	2.44	3.92	<i>0.0765</i>	0.603

3.3 Benchmark: CHARTQA

Table 32: **Calibration Benchmark on docvqa (M3-LLaVA 7B)**. **Bold**: best; *italic*: second best. ↓/↑: lower/higher is better.

Calibration Method	Regime / Sampling	ECE (%) ↓	Ada-ECE (%) ↓	Brier ↓	AUROC ↑
Naive Confidence (NC)	Single-Pass ($T = 0.0$)	74.57	74.57	0.6430	<i>0.845</i>
Temperature Scaling (TS)	Single-Pass ($T = 0.0$)	48.89	48.89	0.2796	<i>0.845</i>
Adaptive TS (ATS)	Single-Pass ($T = 0.0$)	48.87	48.87	0.2794	<i>0.845</i>
Platt Scaling (1D)	Single-Pass ($T = 0.0$)	<i>2.62</i>	<i>2.52</i>	<i>0.0405</i>	<i>0.845</i>
Trajectory Platt (5D)	Single-Pass ($T = 0.0$)	1.87	1.81	0.0377	0.888

Table 33: **Calibration Benchmark on docvqa (MQT-LLaVA 7B)**. **Bold**: best; *italic*: second best. ↓/↑: lower/higher is better.

Calibration Method	Regime / Sampling	ECE (%) ↓	Ada-ECE (%) ↓	Brier ↓	AUROC ↑
Naive Confidence (NC)	Single-Pass ($T = 0.0$)	53.30	53.30	0.4162	<i>0.517</i>
Temperature Scaling (TS)	Single-Pass ($T = 0.0$)	47.07	47.07	0.2584	<i>0.517</i>
Adaptive TS (ATS)	Single-Pass ($T = 0.0$)	47.06	47.06	0.2583	<i>0.517</i>
Platt Scaling (1D)	Single-Pass ($T = 0.0$)	0.09	<i>3.67</i>	0.0362	<i>0.517</i>
Trajectory Platt (5D)	Single-Pass ($T = 0.0$)	<i>1.08</i>	2.06	<i>0.0364</i>	0.594

3.4 Benchmark: DOCVQA

Table 34: **Calibration Benchmark on gqa (M3-LLaVA 7B).** Bold: best; *italic*: second best. ↓/↑: lower/higher is better.

Calibration Method	Regime / Sampling	ECE (%) ↓	Ada-ECE (%) ↓	Brier ↓	AUROC ↑
Naive Confidence (NC)	Single-Pass ($T = 0.0$)	32.55	32.55	0.3223	0.683
Temperature Scaling (TS)	Single-Pass ($T = 0.0$)	8.41	10.85	0.2082	0.683
Adaptive TS (ATS)	Single-Pass ($T = 0.0$)	<i>7.47</i>	11.25	0.2053	0.709
Platt Scaling (1D)	Single-Pass ($T = 0.0$)	5.90	<i>6.89</i>	<i>0.1991</i>	0.683
Trajectory Platt (5D)	Single-Pass ($T = 0.0$)	7.61	6.50	0.1978	<i>0.707</i>

Table 35: **Calibration Benchmark on gqa (MQT-LLaVA 7B).** Bold: best; *italic*: second best. ↓/↑: lower/higher is better.

Calibration Method	Regime / Sampling	ECE (%) ↓	Ada-ECE (%) ↓	Brier ↓	AUROC ↑
Naive Confidence (NC)	Single-Pass ($T = 0.0$)	13.64	10.19	0.1865	0.757
Temperature Scaling (TS)	Single-Pass ($T = 0.0$)	<i>9.21</i>	10.61	0.1860	0.757
Adaptive TS (ATS)	Single-Pass ($T = 0.0$)	7.58	11.99	<i>0.1857</i>	<i>0.759</i>
Platt Scaling (1D)	Single-Pass ($T = 0.0$)	9.56	<i>10.60</i>	0.1861	0.757
Trajectory Platt (5D)	Single-Pass ($T = 0.0$)	13.81	14.66	0.1854	0.763

3.5 Benchmark: GQA

Table 36: **Calibration Benchmark on infographicvqa (M3-LLaVA 7B)**. **Bold**: best; *italic*: second best. ↓/↑: lower/higher is better.

Calibration Method	Regime / Sampling	ECE (%) ↓	Ada-ECE (%) ↓	Brier ↓	AUROC ↑
Naive Confidence (NC)	Single-Pass ($T = 0.0$)	85.32	85.32	0.7761	0.333
Temperature Scaling (TS)	Single-Pass ($T = 0.0$)	54.00	54.00	0.3006	0.333
Adaptive TS (ATS)	Single-Pass ($T = 0.0$)	53.98	53.98	0.3004	0.333
Platt Scaling (1D)	Single-Pass ($T = 0.0$)	<i>0.03</i>	<i>1.22</i>	0.0074	0.667
Trajectory Platt (5D)	Single-Pass ($T = 0.0$)	0.01	1.20	<i>0.0075</i>	<i>0.501</i>

Table 37: **Calibration Benchmark on infographicvqa (MQT-LLaVA 7B)**. **Bold**: best; *italic*: second best. ↓/↑: lower/higher is better.

Calibration Method	Regime / Sampling	ECE (%) ↓	Ada-ECE (%) ↓	Brier ↓	AUROC ↑
Naive Confidence (NC)	Single-Pass ($T = 0.0$)	75.51	75.51	0.6535	<i>0.769</i>
Temperature Scaling (TS)	Single-Pass ($T = 0.0$)	52.63	52.63	0.2826	<i>0.769</i>
Adaptive TS (ATS)	Single-Pass ($T = 0.0$)	52.61	52.61	0.2825	<i>0.769</i>
Platt Scaling (1D)	Single-Pass ($T = 0.0$)	0.28	<i>1.11</i>	<i>0.0050</i>	<i>0.769</i>
Trajectory Platt (5D)	Single-Pass ($T = 0.0$)	<i>0.30</i>	1.08	0.0049	0.930

3.6 Benchmark: INFOGRAPHICVQA

Table 38: **Calibration Benchmark on lego-puzzles (M3-LLaVA 7B)**. **Bold**: best; *italic*: second best. ↓/↑: lower/higher is better.

Calibration Method	Regime / Sampling	ECE (%) ↓	Ada-ECE (%) ↓	Brier ↓	AUROC ↑
Naive Confidence (NC)	Single-Pass ($T = 0.0$)	73.16	73.16	0.7314	<i>0.477</i>
Temperature Scaling (TS)	Single-Pass ($T = 0.0$)	34.71	34.71	0.3160	<i>0.477</i>
Adaptive TS (ATS)	Single-Pass ($T = 0.0$)	34.66	34.66	0.3156	<i>0.477</i>
Platt Scaling (1D)	Single-Pass ($T = 0.0$)	0.71	<i>11.89</i>	<i>0.1958</i>	<i>0.477</i>
Trajectory Platt (5D)	Single-Pass ($T = 0.0$)	<i>4.15</i>	9.40	0.1914	0.601

Table 39: **Calibration Benchmark on lego-puzzles (MQT-LLaVA 7B)**. **Bold**: best; *italic*: second best. ↓/↑: lower/higher is better.

Calibration Method	Regime / Sampling	ECE (%) ↓	Ada-ECE (%) ↓	Brier ↓	AUROC ↑
Naive Confidence (NC)	Single-Pass ($T = 0.0$)	34.60	33.80	0.3474	0.510
Temperature Scaling (TS)	Single-Pass ($T = 0.0$)	23.68	23.68	0.2539	0.510
Adaptive TS (ATS)	Single-Pass ($T = 0.0$)	23.68	23.68	0.2539	<i>0.511</i>
Platt Scaling (1D)	Single-Pass ($T = 0.0$)	0.60	<i>12.43</i>	<i>0.1968</i>	0.490
Trajectory Platt (5D)	Single-Pass ($T = 0.0$)	<i>7.81</i>	8.67	0.1924	0.593

3.7 Benchmark: LEGOPUZZLES

Table 40: **Calibration Benchmark on mmbench (M3-LLaVA 7B)**. **Bold**: best; *italic*: second best. ↓/↑: lower/higher is better.

Calibration Method	Regime / Sampling	ECE (%) ↓	Ada-ECE (%) ↓	Brier ↓	AUROC ↑
Naive Confidence (NC)	Single-Pass ($T = 0.0$)	27.00	27.00	0.2700	0.608
Temperature Scaling (TS)	Single-Pass ($T = 0.0$)	6.43	7.79	0.1956	0.608
Adaptive TS (ATS)	Single-Pass ($T = 0.0$)	2.86	7.37	<i>0.1911</i>	<i>0.655</i>
Platt Scaling (1D)	Single-Pass ($T = 0.0$)	<i>4.60</i>	<i>6.94</i>	0.1970	0.608
Trajectory Platt (5D)	Single-Pass ($T = 0.0$)	6.62	6.54	0.1890	0.660

Table 41: **Calibration Benchmark on mmbench (MQT-LLaVA 7B)**. **Bold**: best; *italic*: second best. ↓/↑: lower/higher is better.

Calibration Method	Regime / Sampling	ECE (%) ↓	Ada-ECE (%) ↓	Brier ↓	AUROC ↑
Naive Confidence (NC)	Single-Pass ($T = 0.0$)	15.38	15.27	0.1866	<i>0.828</i>
Temperature Scaling (TS)	Single-Pass ($T = 0.0$)	9.81	10.24	0.1560	<i>0.828</i>
Adaptive TS (ATS)	Single-Pass ($T = 0.0$)	<i>9.05</i>	10.14	0.1558	0.829
Platt Scaling (1D)	Single-Pass ($T = 0.0$)	8.11	<i>8.38</i>	0.1503	<i>0.828</i>
Trajectory Platt (5D)	Single-Pass ($T = 0.0$)	10.42	7.51	<i>0.1543</i>	0.819

3.8 Benchmark: MMBENCH

Table 42: **Calibration Benchmark on mmmu (M3-LLaVA 7B)**. **Bold**: best; *italic*: second best. ↓/↑: lower/higher is better.

Calibration Method	Regime / Sampling	ECE (%) ↓	Ada-ECE (%) ↓	Brier ↓	AUROC ↑
Naive Confidence (NC)	Single-Pass ($T = 0.0$)	78.76	78.76	0.6889	0.639
Temperature Scaling (TS)	Single-Pass ($T = 0.0$)	52.21	52.21	0.2908	0.639
Adaptive TS (ATS)	Single-Pass ($T = 0.0$)	52.19	52.19	0.2906	0.639
Platt Scaling (1D)	Single-Pass ($T = 0.0$)	<i>1.81</i>	3.44	<i>0.0205</i>	0.639
Trajectory Platt (5D)	Single-Pass ($T = 0.0$)	1.68	<i>3.56</i>	0.0198	<i>0.623</i>

Table 43: **Calibration Benchmark on mmmu (MQT-LLaVA 7B)**. **Bold**: best; *italic*: second best. ↓/↑: lower/higher is better.

Calibration Method	Regime / Sampling	ECE (%) ↓	Ada-ECE (%) ↓	Brier ↓	AUROC ↑
Naive Confidence (NC)	Single-Pass ($T = 0.0$)	52.24	52.24	0.3669	<i>0.315</i>
Temperature Scaling (TS)	Single-Pass ($T = 0.0$)	49.32	49.32	0.2556	<i>0.315</i>
Adaptive TS (ATS)	Single-Pass ($T = 0.0$)	49.31	49.31	0.2555	<i>0.315</i>
Platt Scaling (1D)	Single-Pass ($T = 0.0$)	0.16	<i>2.14</i>	0.0115	<i>0.315</i>
Trajectory Platt (5D)	Single-Pass ($T = 0.0$)	<i>0.23</i>	2.04	<i>0.0116</i>	0.724

3.9 Benchmark: MMMU

Table 44: **Calibration Benchmark on pope (M3-LLaVA 7B)**. **Bold**: best; *italic*: second best. ↓/↑: lower/higher is better.

Calibration Method	Regime / Sampling	ECE (%) ↓	Ada-ECE (%) ↓	Brier ↓	AUROC ↑
Naive Confidence (NC)	Single-Pass ($T = 0.0$)	3.45	4.40	<i>0.0882</i>	0.739
Temperature Scaling (TS)	Single-Pass ($T = 0.0$)	4.27	4.37	0.0883	0.739
Adaptive TS (ATS)	Single-Pass ($T = 0.0$)	4.42	4.27	0.0879	<i>0.750</i>
Platt Scaling (1D)	Single-Pass ($T = 0.0$)	<i>4.19</i>	<i>4.32</i>	<i>0.0882</i>	0.739
Trajectory Platt (5D)	Single-Pass ($T = 0.0$)	4.82	6.62	0.0887	0.751

Table 45: **Calibration Benchmark on pope (MQT-LLaVA 7B)**. **Bold**: best; *italic*: second best. ↓/↑: lower/higher is better.

Calibration Method	Regime / Sampling	ECE (%) ↓	Ada-ECE (%) ↓	Brier ↓	AUROC ↑
Naive Confidence (NC)	Single-Pass ($T = 0.0$)	4.01	4.30	0.0903	0.876
Temperature Scaling (TS)	Single-Pass ($T = 0.0$)	3.52	4.93	<i>0.0902</i>	0.876
Adaptive TS (ATS)	Single-Pass ($T = 0.0$)	3.75	4.90	<i>0.0902</i>	<i>0.874</i>
Platt Scaling (1D)	Single-Pass ($T = 0.0$)	3.97	4.64	0.0893	0.876
Trajectory Platt (5D)	Single-Pass ($T = 0.0$)	<i>3.55</i>	<i>4.48</i>	0.0903	0.873

3.10 Benchmark: POPE

Table 46: **Calibration Benchmark on scienceqa (M3-LLaVA 7B)**. **Bold**: best; *italic*: second best. ↓/↑: lower/higher is better.

Calibration Method	Regime / Sampling	ECE (%) ↓	Ada-ECE (%) ↓	Brier ↓	AUROC ↑
Naive Confidence (NC)	Single-Pass ($T = 0.0$)	22.16	21.93	0.2368	0.807
Temperature Scaling (TS)	Single-Pass ($T = 0.0$)	13.10	12.68	0.1924	0.807
Adaptive TS (ATS)	Single-Pass ($T = 0.0$)	12.98	13.00	0.1918	<i>0.808</i>
Platt Scaling (1D)	Single-Pass ($T = 0.0$)	4.80	4.99	<i>0.1732</i>	0.807
LN-Entropy	Multi-Pass ($T = 0.5, K = 10$)	-	-	-	-
Semantic Entropy	Multi-Pass ($T = 0.5, K = 10$)	-	-	-	-
EigenScore	Multi-Pass ($T = 0.5, K = 10$)	-	-	-	-
UMPIRE	Multi-Pass ($T = 0.5, K = 10$)	-	-	-	-
Trajectory Platt (5D)	Single-Pass ($T = 0.0$)	<i>5.17</i>	<i>6.15</i>	0.1719	0.813

Table 47: **Calibration Benchmark on scienceqa (MQT-LLaVA 7B)**. **Bold**: best; *italic*: second best. ↓/↑: lower/higher is better.

Calibration Method	Regime / Sampling	ECE (%) ↓	Ada-ECE (%) ↓	Brier ↓	AUROC ↑
Naive Confidence (NC)	Single-Pass ($T = 0.0$)	23.65	23.17	0.2579	0.760
Temperature Scaling (TS)	Single-Pass ($T = 0.0$)	9.32	10.23	0.1975	0.760
Adaptive TS (ATS)	Single-Pass ($T = 0.0$)	8.73	9.38	0.1966	<i>0.761</i>
Platt Scaling (1D)	Single-Pass ($T = 0.0$)	<i>7.99</i>	<i>6.91</i>	<i>0.1916</i>	0.760
LN-Entropy	Multi-Pass ($T = 0.5, K = 10$)	-	-	-	-
Semantic Entropy	Multi-Pass ($T = 0.5, K = 10$)	-	-	-	-
EigenScore	Multi-Pass ($T = 0.5, K = 10$)	-	-	-	-
UMPIRE	Multi-Pass ($T = 0.5, K = 10$)	-	-	-	-
Trajectory Platt (5D)	Single-Pass ($T = 0.0$)	6.46	5.72	0.1860	0.769

3.11 Benchmark: SCIENCEQA

Table 48: **Calibration Benchmark on seedbench (M3-LLaVA 7B)**. **Bold**: best; *italic*: second best. ↓/↑: lower/higher is better.

Calibration Method	Regime / Sampling	ECE (%) ↓	Ada-ECE (%) ↓	Brier ↓	AUROC ↑
Naive Confidence (NC)	Single-Pass ($T = 0.0$)	32.25	32.25	0.3225	0.483
Temperature Scaling (TS)	Single-Pass ($T = 0.0$)	<i>0.66</i>	<i>5.25</i>	0.2185	0.483
Adaptive TS (ATS)	Single-Pass ($T = 0.0$)	0.87	8.60	<i>0.2180</i>	<i>0.554</i>
Platt Scaling (1D)	Single-Pass ($T = 0.0$)	0.48	5.20	0.2184	0.483
Trajectory Platt (5D)	Single-Pass ($T = 0.0$)	3.94	7.68	0.2071	0.642

Table 49: **Calibration Benchmark on seedbench (MQT-LLaVA 7B)**. **Bold**: best; *italic*: second best. ↓/↑: lower/higher is better.

Calibration Method	Regime / Sampling	ECE (%) ↓	Ada-ECE (%) ↓	Brier ↓	AUROC ↑
Naive Confidence (NC)	Single-Pass ($T = 0.0$)	18.56	17.74	0.2297	0.733
Temperature Scaling (TS)	Single-Pass ($T = 0.0$)	6.68	10.44	0.2015	0.733
Adaptive TS (ATS)	Single-Pass ($T = 0.0$)	6.04	<i>9.65</i>	0.2009	<i>0.736</i>
Platt Scaling (1D)	Single-Pass ($T = 0.0$)	9.51	8.62	<i>0.1986</i>	0.733
Trajectory Platt (5D)	Single-Pass ($T = 0.0$)	<i>6.66</i>	9.67	0.1957	0.741

3.12 Benchmark: SEEDBENCH

Table 50: **Calibration Benchmark on textvqa (M3-LLaVA 7B)**. **Bold**: best; *italic*: second best. ↓/↑: lower/higher is better.

Calibration Method	Regime / Sampling	ECE (%) ↓	Ada-ECE (%) ↓	Brier ↓	AUROC ↑
Naive Confidence (NC)	Single-Pass ($T = 0.0$)	10.50	11.07	0.1714	0.839
Temperature Scaling (TS)	Single-Pass ($T = 0.0$)	10.38	11.14	0.1712	0.839
Adaptive TS (ATS)	Single-Pass ($T = 0.0$)	10.15	11.31	0.1710	0.839
Platt Scaling (1D)	Single-Pass ($T = 0.0$)	<i>8.47</i>	<i>8.22</i>	0.1615	0.839
LN-Entropy	Multi-Pass ($T = 0.5, K = 10$)	-	-	-	-
Semantic Entropy	Multi-Pass ($T = 0.5, K = 10$)	-	-	-	-
EigenScore	Multi-Pass ($T = 0.5, K = 10$)	-	-	-	-
UMPIRE	Multi-Pass ($T = 0.5, K = 10$)	-	-	-	-
Trajectory Platt (5D)	Single-Pass ($T = 0.0$)	5.78	5.35	<i>0.1628</i>	<i>0.835</i>

Table 51: **Calibration Benchmark on textvqa (MQT-LLaVA 7B)**. **Bold**: best; *italic*: second best. ↓/↑: lower/higher is better.

Calibration Method	Regime / Sampling	ECE (%) ↓	Ada-ECE (%) ↓	Brier ↓	AUROC ↑
Naive Confidence (NC)	Single-Pass ($T = 0.0$)	32.27	31.91	0.3286	0.684
Temperature Scaling (TS)	Single-Pass ($T = 0.0$)	18.21	18.32	0.2503	0.684
Adaptive TS (ATS)	Single-Pass ($T = 0.0$)	18.36	18.36	0.2500	<i>0.689</i>
Platt Scaling (1D)	Single-Pass ($T = 0.0$)	<i>8.95</i>	<i>10.46</i>	<i>0.2148</i>	0.684
LN-Entropy	Multi-Pass ($T = 0.5, K = 10$)	-	-	-	-
Semantic Entropy	Multi-Pass ($T = 0.5, K = 10$)	-	-	-	-
EigenScore	Multi-Pass ($T = 0.5, K = 10$)	-	-	-	-
UMPIRE	Multi-Pass ($T = 0.5, K = 10$)	-	-	-	-
Trajectory Platt (5D)	Single-Pass ($T = 0.0$)	5.25	7.76	0.1948	0.719

3.13 Benchmark: TEXTVQA

Table 52: **Calibration Benchmark on vizwiz-vqa (M3-LLaVA 7B).** Bold: best; *italic*: second best. ↓/↑: lower/higher is better.

Calibration Method	Regime / Sampling	ECE (%) ↓	Ada-ECE (%) ↓	Brier ↓	AUROC ↑
Naive Confidence (NC)	Single-Pass ($T = 0.0$)	15.97	15.97	0.2071	<i>0.793</i>
Temperature Scaling (TS)	Single-Pass ($T = 0.0$)	15.09	15.09	0.2065	<i>0.793</i>
Adaptive TS (ATS)	Single-Pass ($T = 0.0$)	14.87	14.87	<i>0.2058</i>	<i>0.793</i>
Platt Scaling (1D)	Single-Pass ($T = 0.0$)	<i>6.64</i>	5.80	0.1828	<i>0.793</i>
LN-Entropy	Multi-Pass ($T = 0.5, K = 10$)	-	-	-	-
Semantic Entropy	Multi-Pass ($T = 0.5, K = 10$)	-	-	-	-
EigenScore	Multi-Pass ($T = 0.5, K = 10$)	-	-	-	-
UMPIRE	Multi-Pass ($T = 0.5, K = 10$)	-	-	-	-
Trajectory Platt (5D)	Single-Pass ($T = 0.0$)	6.38	<i>6.57</i>	0.1828	0.794

Table 53: **Calibration Benchmark on vizwiz-vqa (MQT-LLaVA 7B).** Bold: best; *italic*: second best. ↓/↑: lower/higher is better.

Calibration Method	Regime / Sampling	ECE (%) ↓	Ada-ECE (%) ↓	Brier ↓	AUROC ↑
Naive Confidence (NC)	Single-Pass ($T = 0.0$)	30.66	30.66	0.2435	0.797
Temperature Scaling (TS)	Single-Pass ($T = 0.0$)	29.41	29.41	0.2328	0.797
Adaptive TS (ATS)	Single-Pass ($T = 0.0$)	29.11	29.11	0.2312	<i>0.796</i>
Platt Scaling (1D)	Single-Pass ($T = 0.0$)	<i>6.27</i>	7.62	<i>0.1433</i>	0.797
LN-Entropy	Multi-Pass ($T = 0.5, K = 10$)	-	-	-	-
Semantic Entropy	Multi-Pass ($T = 0.5, K = 10$)	-	-	-	-
EigenScore	Multi-Pass ($T = 0.5, K = 10$)	-	-	-	-
UMPIRE	Multi-Pass ($T = 0.5, K = 10$)	-	-	-	-
Trajectory Platt (5D)	Single-Pass ($T = 0.0$)	6.03	<i>8.17</i>	0.1418	0.795

3.14 Benchmark: VIZWIZVQA

Table 54: **Calibration Benchmark on vllm-safety (M3-LLaVA 7B).** Bold: best; *italic*: second best. ↓/↑: lower/higher is better.

Calibration Method	Regime / Sampling	ECE (%) ↓	Ada-ECE (%) ↓	Brier ↓	AUROC ↑
Naive Confidence (NC)	Single-Pass ($T = 0.0$)	<i>11.59</i>	<i>10.50</i>	0.1732	0.814
Temperature Scaling (TS)	Single-Pass ($T = 0.0$)	16.98	16.80	0.1812	0.814
Adaptive TS (ATS)	Single-Pass ($T = 0.0$)	14.08	13.84	<i>0.1730</i>	<i>0.813</i>
Platt Scaling (1D)	Single-Pass ($T = 0.0$)	14.76	14.04	0.1798	0.814
LN-Entropy	Multi-Pass ($T = 0.5, K = 10$)	-	-	-	-
Semantic Entropy	Multi-Pass ($T = 0.5, K = 10$)	-	-	-	-
EigenScore	Multi-Pass ($T = 0.5, K = 10$)	-	-	-	-
UMPIRE	Multi-Pass ($T = 0.5, K = 10$)	-	-	-	-
Trajectory Platt (5D)	Single-Pass ($T = 0.0$)	5.87	6.88	0.1709	0.773

Table 55: **Calibration Benchmark on vllm-safety (MQT-LLaVA 7B).** Bold: best; *italic*: second best. ↓/↑: lower/higher is better.

Calibration Method	Regime / Sampling	ECE (%) ↓	Ada-ECE (%) ↓	Brier ↓	AUROC ↑
Naive Confidence (NC)	Single-Pass ($T = 0.0$)	<i>9.63</i>	<i>9.74</i>	0.1966	<i>0.763</i>
Temperature Scaling (TS)	Single-Pass ($T = 0.0$)	14.20	14.42	0.2054	<i>0.763</i>
Adaptive TS (ATS)	Single-Pass ($T = 0.0$)	12.13	13.19	0.1966	0.768
Platt Scaling (1D)	Single-Pass ($T = 0.0$)	11.34	11.29	<i>0.1937</i>	<i>0.763</i>
LN-Entropy	Multi-Pass ($T = 0.5, K = 10$)	-	-	-	-
Semantic Entropy	Multi-Pass ($T = 0.5, K = 10$)	-	-	-	-
EigenScore	Multi-Pass ($T = 0.5, K = 10$)	-	-	-	-
UMPIRE	Multi-Pass ($T = 0.5, K = 10$)	-	-	-	-
Trajectory Platt (5D)	Single-Pass ($T = 0.0$)	5.68	6.58	0.1921	0.733

3.15 Benchmark: VLLM-SAFETY

Table 56: **Calibration Benchmark on vqav2 (M3-LLaVA 7B)**. **Bold**: best; *italic*: second best. ↓/↑: lower/higher is better.

Calibration Method	Regime / Sampling	ECE (%) ↓	Ada-ECE (%) ↓	Brier ↓	AUROC ↑
Naive Confidence (NC)	Single-Pass ($T = 0.0$)	8.64	<i>7.23</i>	0.1387	<i>0.772</i>
Temperature Scaling (TS)	Single-Pass ($T = 0.0$)	9.40	8.58	0.1417	<i>0.772</i>
Adaptive TS (ATS)	Single-Pass ($T = 0.0$)	8.39	<i>7.52</i>	<i>0.1381</i>	0.788
Platt Scaling (1D)	Single-Pass ($T = 0.0$)	5.52	<i>6.97</i>	0.1391	<i>0.772</i>
LN-Entropy	Multi-Pass ($T = 0.5, K = 10$)	-	-	-	-
Semantic Entropy	Multi-Pass ($T = 0.5, K = 10$)	-	-	-	-
EigenScore	Multi-Pass ($T = 0.5, K = 10$)	-	-	-	-
UMPIRE	Multi-Pass ($T = 0.5, K = 10$)	-	-	-	-
Trajectory Platt (5D)	Single-Pass ($T = 0.0$)	<i>6.86</i>	6.50	0.1380	0.788

Table 57: **Calibration Benchmark on vqav2 (MQT-LLaVA 7B)**. **Bold**: best; *italic*: second best. ↓/↑: lower/higher is better.

Calibration Method	Regime / Sampling	ECE (%) ↓	Ada-ECE (%) ↓	Brier ↓	AUROC ↑
Naive Confidence (NC)	Single-Pass ($T = 0.0$)	<i>10.49</i>	<i>10.31</i>	0.1713	0.641
Temperature Scaling (TS)	Single-Pass ($T = 0.0$)	13.87	13.65	0.1789	0.641
Adaptive TS (ATS)	Single-Pass ($T = 0.0$)	12.36	12.03	0.1753	<i>0.658</i>
Platt Scaling (1D)	Single-Pass ($T = 0.0$)	10.54	10.94	<i>0.1712</i>	0.641
LN-Entropy	Multi-Pass ($T = 0.5, K = 10$)	-	-	-	-
Semantic Entropy	Multi-Pass ($T = 0.5, K = 10$)	-	-	-	-
EigenScore	Multi-Pass ($T = 0.5, K = 10$)	-	-	-	-
UMPIRE	Multi-Pass ($T = 0.5, K = 10$)	-	-	-	-
Trajectory Platt (5D)	Single-Pass ($T = 0.0$)	7.16	7.64	0.1629	0.739

3.16 Benchmark: VQAV2

Table 58: **Combinatorial 5D Trajectory Feature Cardinality Progression** ($k \in \{1 \dots 5\}$). Macro-averaged out-of-fold calibration error and discrimination across Core 5 vision-language benchmarks for M3-LLaVA (7B) and MQT-LLaVA (7B). Evaluates all $\sum_{k=1}^5 \binom{5}{k} = 31$ feature subsets. Reported in pure decimal format. **Bold**: Rank 1, *Italic*: Rank 2 within each architecture. Note: Subsets lacking the base logit anchor x_1 achieve artificially low quantile Ada-ECE via class-prior probability clustering, but suffer severe AUROC collapse (≤ 0.64 vs. $0.71+$ for x_1 -anchored models); joint calibration and discrimination require the full 5D representation.

Cardinality (k)	Subsets ($\binom{5}{k}$)	Optimal Formula	Macro Ada-ECE (Best) $\downarrow$	Macro Ada-ECE (Mean $\pm$)
M3-LLaVA (7B)				
$k = 1$	5	x_5	0.0323	0.0430 ± 0.0108
$k = 2$	10	$x_2 + x_4$	<i>0.0314</i>	0.0435 ± 0.0078
$k = 3$	10	$x_2 + x_4 + x_5$	0.0290	0.0416 ± 0.0057
$k = 4$	5	$x_1 + x_3 + x_4 + x_5$	0.0356	0.0391 ± 0.0024
$k = 5$	1	$x_1 + x_2 + x_3 + x_4 + x_5$	0.0365	0.0365 ± 0.0000
MQT-LLaVA (7B)				
$k = 1$	5	x_5	0.0297	0.0461 ± 0.0154
$k = 2$	10	$x_4 + x_5$	<i>0.0268</i>	0.0448 ± 0.0139
$k = 3$	10	$x_2 + x_4 + x_5$	0.0235	0.0397 ± 0.0091
$k = 4$	5	$x_1 + x_2 + x_3 + x_4$	0.0341	0.0368 ± 0.0029
$k = 5$	1	$x_1 + x_2 + x_3 + x_4 + x_5$	0.0344	0.0344 ± 0.0000

4 Group 8: Combinatorial 5D Trajectory Feature Ablation Study

Exhaustive 5-fold cross-validation combinatorial ablation across all $\sum_{k=1}^5 \binom{5}{k} = 31$ feature subsets of canonical 5D trajectory features across all 14 vision-language benchmarks on M3-LLaVA (7B) and MQT-LLaVA (7B).

4.1 Table 4: Macro Cardinality Progression ($k \in \{1 \dots 5\}$)

Table 59: **Leave-One-Out (LOO) Feature Degradation Sensitivity across Core 5 Benchmarks.** Marginal change in calibration error ($\Delta\text{Ada-ECE} = \text{Ada-ECE}_{-j} - \text{Ada-ECE}_{\text{Full 5D}}$) and discrimination ($\Delta\text{AUROC} = \text{AUROC}_{-j} - \text{AUROC}_{\text{Full 5D}}$) when dropping individual feature x_j from Trajectory Platt (5D). Positive $\Delta\text{Ada-ECE}$ and negative ΔAUROC indicate feature necessity. **Bold:** Rank 1, *Italic:* Rank 2 degradation impact.

Dropped Feature	Functional Role	M3-LLaVA (7B)		MQT-I
		Δ Ada-ECE	Δ AUROC	Δ Ada-EC
x_1 (Final Logit)	Base Fine-Scale Log-Odds Anchor ($\ell_{576/256}$)	<i>+0.0038</i>	-0.094	+0.0075
x_2 (Discrete Answer Stability)	Categorical Scale Consistency	-0.0009	-0.001	+0.0014
x_3 (Scale Entropy Slope)	Predictive Entropy Decay Dynamics	+0.0063	<i>-0.002</i>	<i>+0.0035</i>
x_4 (Answer Flip Frequency)	Argmax Scale Transition Rate	+0.0014	-0.001	+0.0000
x_5 (Monotonicity Count)	Directional Monotonicity Trend	+0.0023	+0.000	-0.0003

4.2 Table 5: Leave-One-Out (LOO) Feature Degradation Sensitivity

Table 60: **Exhaustive 14-Benchmark Progression Grid: Adaptive ECE ($\downarrow$)**. Per-dataset comparison across 1D Platt ($\{x_1\}$), Optimal 2D, Optimal 3D, Optimal 4D, and Full Trajectory Platt 5D. Option A ranking: **Bold** Rank 1, *Italic* Rank 2 across each benchmark row.

Benchmark Dataset	1D Platt ($\{x_1\}$)	Best-2D	Best-3D	Best-4D	Full 5D (Ours)
M3-LLaVA (7B)					
AI2D	0.0418	0.0311	<i>0.0335</i>	0.0440	0.0620
ChartQA	0.0149	0.0124	<i>0.0129</i>	0.0139	0.0153
DocVQA	0.0115	<i>0.0085</i>	0.0088	0.0083	0.0121
GQA	0.0411	<i>0.0255</i>	0.0243	0.0261	0.0332
InfoVQA	0.0053	0.0044	<i>0.0047</i>	0.0056	0.0074
LegoPuzzles	0.0949	0.0459	<i>0.0357</i>	0.0413	0.0345
MMBench	0.0419	0.0256	<i>0.0271</i>	0.0355	0.0420
MMMU	0.0131	0.0092	<i>0.0098</i>	0.0122	0.0155
POPE	0.0137	<i>0.0116</i>	0.0104	0.0135	0.0185
ScienceQA	0.0407	0.0208	<i>0.0292</i>	0.0370	0.0351
SEED-Bench	0.0274	<i>0.0259</i>	0.0269	0.0263	0.0230
TextVQA	0.0432	<i>0.0226</i>	0.0166	0.0359	0.0376
VizWiz-VQA	0.0259	0.0172	0.0202	<i>0.0188</i>	0.0204
VQAv2	0.0764	0.0204	<i>0.0206</i>	0.0238	0.0272
Macro Mean	0.0351	0.0201	0.0201	<i>0.0244</i>	0.0274
MQT-LLaVA (7B)					
AI2D	0.0362	0.0152	<i>0.0176</i>	0.0294	0.0371
ChartQA	0.0454	0.0165	<i>0.0123</i>	0.0122	0.0179
DocVQA	0.0231	<i>0.0083</i>	0.0077	<i>0.0083</i>	0.0118
GQA	0.0525	0.0287	<i>0.0335</i>	0.0353	0.0544
InfoVQA	0.0065	0.0063	0.0054	0.0046	<i>0.0051</i>
LegoPuzzles	0.0899	0.0395	0.0303	<i>0.0383</i>	0.0520
MMBench	0.0407	0.0321	<i>0.0391</i>	0.0453	0.0490
MMMU	0.0128	0.0098	0.0079	<i>0.0090</i>	0.0116
POPE	0.0236	0.0198	<i>0.0175</i>	0.0171	0.0210
ScienceQA	0.0511	0.0329	0.0282	<i>0.0310</i>	0.0404
SEED-Bench	0.0303	0.0227	<i>0.0299</i>	0.0338	0.0530
TextVQA	0.0997	0.0223	<i>0.0225</i>	0.0386	0.0345
VizWiz-VQA	0.0446	0.0092	<i>0.0237</i>	0.0302	0.0305
VQAv2	0.0880	0.0232	0.0256	<i>0.0255</i>	0.0296
Macro Mean	0.0460	0.0205	<i>0.0215</i>	0.0256	0.0320

4.3 Table 6: Exhaustive 14-Benchmark Adaptive ECE Progression Grid

Table 61: **Combinatorial 5D Trajectory Feature Cardinality Progression on TextVQA** ($k \in \{1 \dots 5\}$). Out-of-fold calibration error and discrimination across feature cardinality levels for M3-LLaVA (7B) and MQT-LLaVA (7B). Evaluated across all $\sum_{k=1}^5 \binom{5}{k} = 31$ feature subsets. Reported in pure decimal format. **Bold**: Rank 1, *Italic*: Rank 2 within each architecture. Note: Subsets lacking the base logit anchor x_1 achieve artificially low quantile Ada-ECE via class-prior probability clustering, but suffer severe AUROC collapse and degraded Brier scores; joint calibration and discrimination require the full 5D representation.

Cardinality (k)	Subsets	Optimal Formula	Ada-ECE ↓	Mean ± Std	AUROC ↑	Brier ↓
M3-LLaVA (7B)						
$k = 1$	5	x_5	0.0297	0.0504 ± 0.0192	0.583	0.2371
$k = 2$	10	$x_4 + x_5$	<i>0.0226</i>	0.0501 ± 0.0144	<i>0.731</i>	0.2039
$k = 3$	10	$x_2 + x_4 + x_5$	0.0166	0.0425 ± 0.0111	0.730	0.2040
$k = 4$	5	$x_1 + x_3 + x_4 + x_5$	0.0359	0.0402 ± 0.0027	0.842	<i>0.1595</i>
$k = 5$	1	$x_1 + x_2 + x_3 + x_4 + x_5$	0.0376	0.0376 ± 0.0000	0.842	0.1593
MQT-LLaVA (7B)						
$k = 1$	5	x_5	0.0223	0.0527 ± 0.0307	0.607	0.2197
$k = 2$	10	$x_2 + x_4$	0.0223	0.0456 ± 0.0149	0.633	0.2163
$k = 3$	10	$x_2 + x_4 + x_5$	<i>0.0225</i>	0.0427 ± 0.0115	0.682	0.2070
$k = 4$	5	$x_1 + x_2 + x_4 + x_5$	0.0386	0.0448 ± 0.0054	<i>0.738</i>	<i>0.1926</i>
$k = 5$	1	$x_1 + x_2 + x_3 + x_4 + x_5$	0.0345	0.0345 ± 0.0000	0.747	0.1851

4.4 Table 7: Cardinality Progression on TextVQA

Table 62: **Combinatorial 5D Trajectory Feature Cardinality Progression on VQAv2** ($k \in \{1 \dots 5\}$). Out-of-fold calibration error and discrimination across feature cardinality levels for M3-LLaVA (7B) and MQT-LLaVA (7B). Evaluated across all $\sum_{k=1}^5 \binom{5}{k} = 31$ feature subsets. Reported in pure decimal format. **Bold**: Rank 1, *Italic*: Rank 2 within each architecture. Note: Subsets lacking the base logit anchor x_1 achieve artificially low quantile Ada-ECE via class-prior probability clustering, but suffer severe AUROC collapse and degraded Brier scores; joint calibration and discrimination require the full 5D representation.

Cardinality (k)	Subsets	Optimal Formula	Ada-ECE ↓	Mean ± Std	AUROC ↑	Brier ↓
M3-LLaVA (7B)						
$k = 1$	5	x_5	0.0271	0.0411 ± 0.0202	0.580	0.1634
$k = 2$	10	$x_4 + x_5$	0.0204	0.0408 ± 0.0169	0.710	0.1481
$k = 3$	10	$x_2 + x_4 + x_5$	<i>0.0206</i>	0.0361 ± 0.0111	0.714	0.1464
$k = 4$	5	$x_2 + x_3 + x_4 + x_5$	0.0238	0.0316 ± 0.0059	<i>0.752</i>	<i>0.1431</i>
$k = 5$	1	$x_1 + x_2 + x_3 + x_4 + x_5$	0.0272	0.0272 ± 0.0000	0.788	0.1362
MQT-LLaVA (7B)						
$k = 1$	5	x_4	0.0259	0.0430 ± 0.0257	0.661	0.1633
$k = 2$	10	$x_4 + x_5$	0.0232	0.0385 ± 0.0140	0.703	0.1611
$k = 3$	10	$x_2 + x_4 + x_5$	0.0256	0.0327 ± 0.0062	0.700	0.1614
$k = 4$	5	$x_1 + x_2 + x_4 + x_5$	<i>0.0255</i>	0.0285 ± 0.0019	<i>0.729</i>	<i>0.1558</i>
$k = 5$	1	$x_1 + x_2 + x_3 + x_4 + x_5$	0.0296	0.0296 ± 0.0000	0.771	0.1511

4.5 Table 8: Cardinality Progression on VQAv2

Table 63: **Combinatorial 5D Trajectory Feature Cardinality Progression on LegoPuzzles** ($k \in \{1 \dots 5\}$). Out-of-fold calibration error and discrimination across feature cardinality levels for M3-LLaVA (7B) and MQT-LLaVA (7B). Evaluated across all $\sum_{k=1}^5 \binom{5}{k} = 31$ feature subsets. Reported in pure decimal format. **Bold**: Rank 1, *Italic*: Rank 2 within each architecture. Note: Subsets lacking the base logit anchor x_1 achieve artificially low quantile Ada-ECE via class-prior probability clustering, but suffer severe AUROC collapse and degraded Brier scores; joint calibration and discrimination require the full 5D representation.

Cardinality (k)	Subsets	Optimal Formula	Ada-ECE $\downarrow$	Mean $\pm$ Std	AUROC $\uparrow$	Brier $\downarrow$
M3-LLaVA (7B)						
$k = 1$	5	x_5	0.0411	0.0581 ± 0.0216	0.516	0.1971
$k = 2$	10	$x_2 + x_4$	0.0459	0.0589 ± 0.0151	0.494	0.1981
$k = 3$	10	$x_3 + x_4 + x_5$	<i>0.0357</i>	0.0568 ± 0.0186	0.589	0.1927
$k = 4$	5	$x_1 + x_2 + x_3 + x_5$	0.0413	0.0495 ± 0.0106	<i>0.583</i>	0.1927
$k = 5$	1	$x_1 + x_2 + x_3 + x_4 + x_5$	0.0345	0.0345 ± 0.0000	0.579	<i>0.1932</i>
MQT-LLaVA (7B)						
$k = 1$	5	x_2	<i>0.0337</i>	0.0549 ± 0.0212	0.485	0.1982
$k = 2$	10	$x_2 + x_4$	0.0395	0.0623 ± 0.0260	0.485	0.2002
$k = 3$	10	$x_1 + x_3 + x_4$	0.0303	0.0535 ± 0.0224	0.613	0.1913
$k = 4$	5	$x_1 + x_2 + x_3 + x_4$	0.0383	0.0536 ± 0.0172	0.607	0.1934
$k = 5$	1	$x_1 + x_2 + x_3 + x_4 + x_5$	0.0520	0.0520 ± 0.0000	<i>0.611</i>	<i>0.1929</i>

4.6 Table 9: Cardinality Progression on LegoPuzzles